Orden Topológico, SCC, 2-SAT

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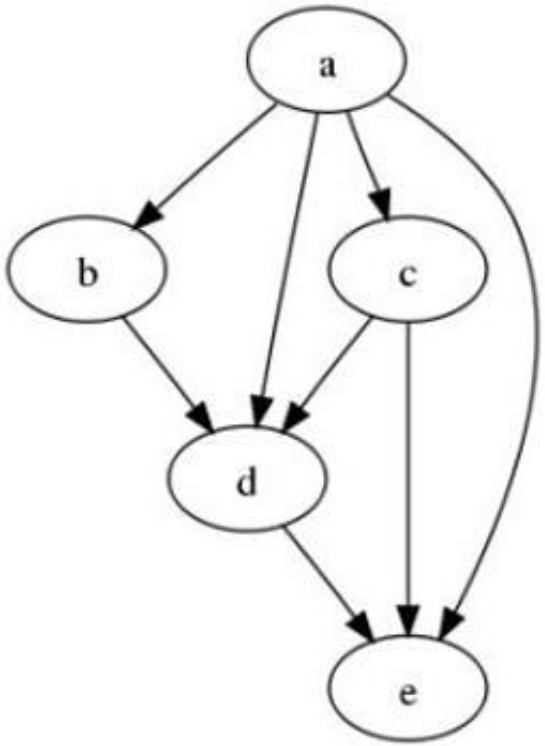
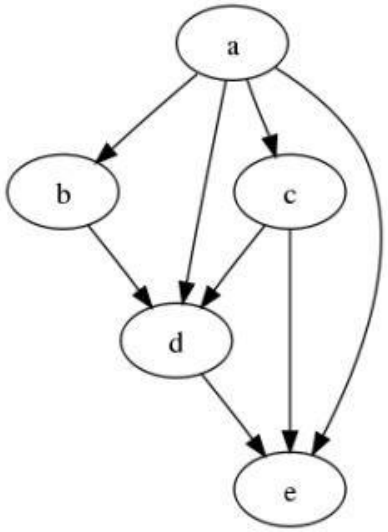
DAG(Directed Acyclic Graph)



Orden Topológico

Dado un grafo dirigido, un orden topológico es un ordenamiento de los nodos de modo que para cada arista x->y, x viene antes que y en el orden.



Este grafo tiene dos órdenes topológicos:

- a,b,c,d,e
- a,c,b,d,e

Orden Topológico

Un grafo tiene orden topológico si y sólo si no tiene ciclo. A estos grafos se los llama DAG (directed-acyclic-graph o “grafo dirigido sin ciclos”).

Para obtener el orden topológico en un DAG:

- Obtenemos el grado de entrada de cada nodo y metemos los de grado 0 en una cola.

- Mientras que la cola no esté vacía, tomamos el nodo del frente y “lo eliminamos” del grafo, actualizando el grado de entrada de sus vecinos y metiendo a la cola los que pasen a tener grado 0.

Orden Topológico

```
vector<int> topSort(vector<vector<int>> &g, vector<int> in)
{
    int n = in.size();
    vector<int> topo;
    queue<int> q;
    for(int i = 0; i < n; i++)
        if(in[i] == 0)
            q.push(i);
    while(!q.empty()) {
        int node = q.front(); q.pop();
        topo.push_back(node);
        for(int x : g[node])
            if(--in[x] == 0)
                q.push(x);
    }
    return topo;
}
```

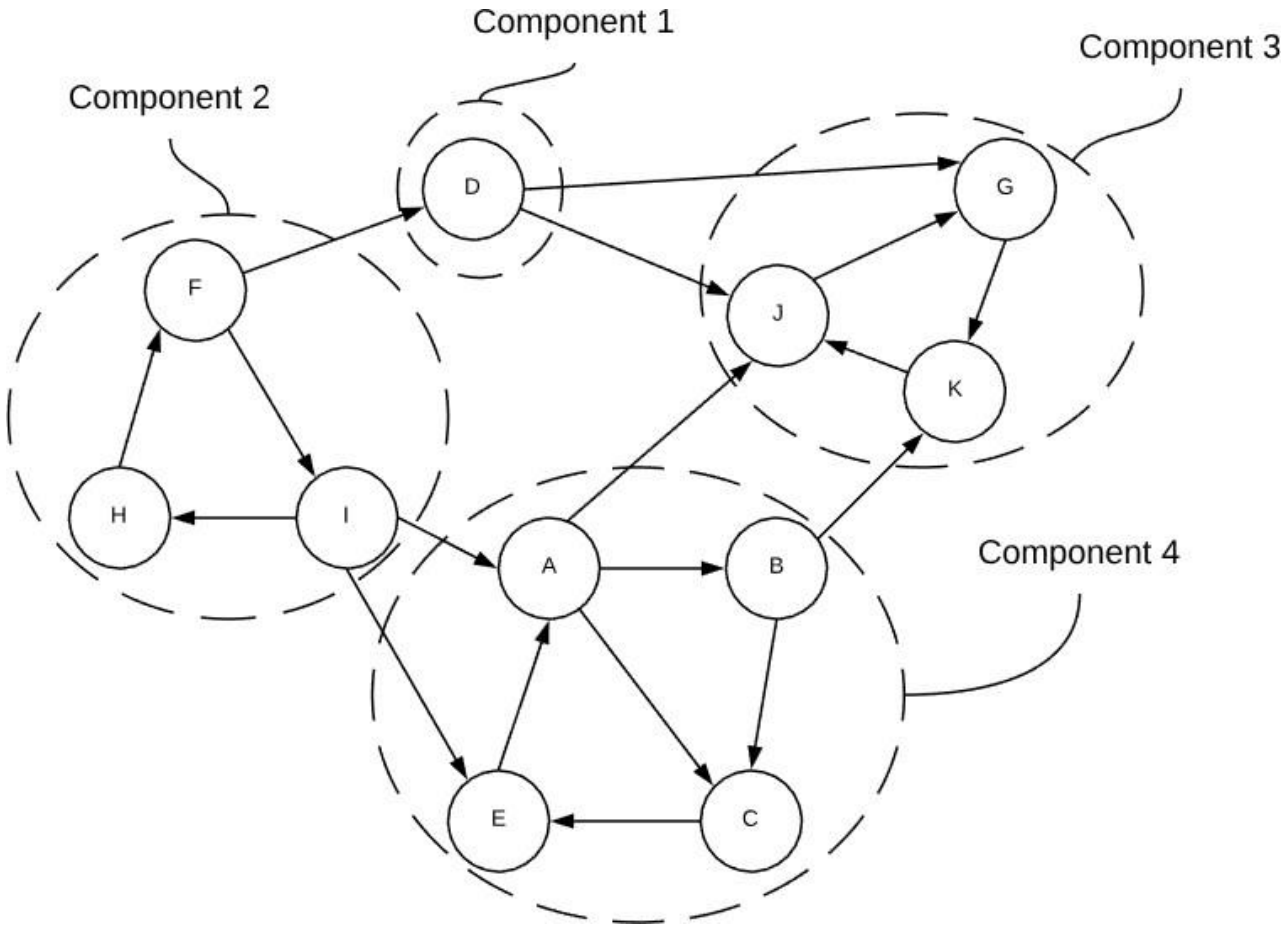
Problema Ejemplo

- https://codeforces.com/contest/1931/problem/F

Componentes fuertemente Conexas(SCC)

- Grafo fuertemente conexo: Un grafo dirigido G es llamado fuertemente conexo si para cada par de vértices u y v existe un camino de u hacia v y un camino de v hacia u.
- Componentes fuertemente conexas (Strongly Connected Components, SCC): Los componentes fuertemente conexos de un grafo dirigido son sus subgrafos maximales fuertemente conexos.

SCC


Component 1
Component 2
Component 3
Component 4
D
G
F
J
K
H
I
A
B
E
C

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- Con un primer DFS, obtenemos la lista de nodos en orden inverso de finalización (sería el orden topológico si fuera un DAG).
- Damos la vuelta todas las aristas de G (obteniendo G’). Esto no cambia las componentes fuertemente conexas, e invierte todas las aristas de la condensación de G.
- Continuamos con DFS partiendo de los nodos en el orden inverso, los demás nodos alcanzables por este formarán parte de la nueva componente fuertemente conexa.

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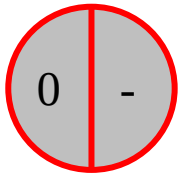
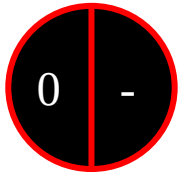
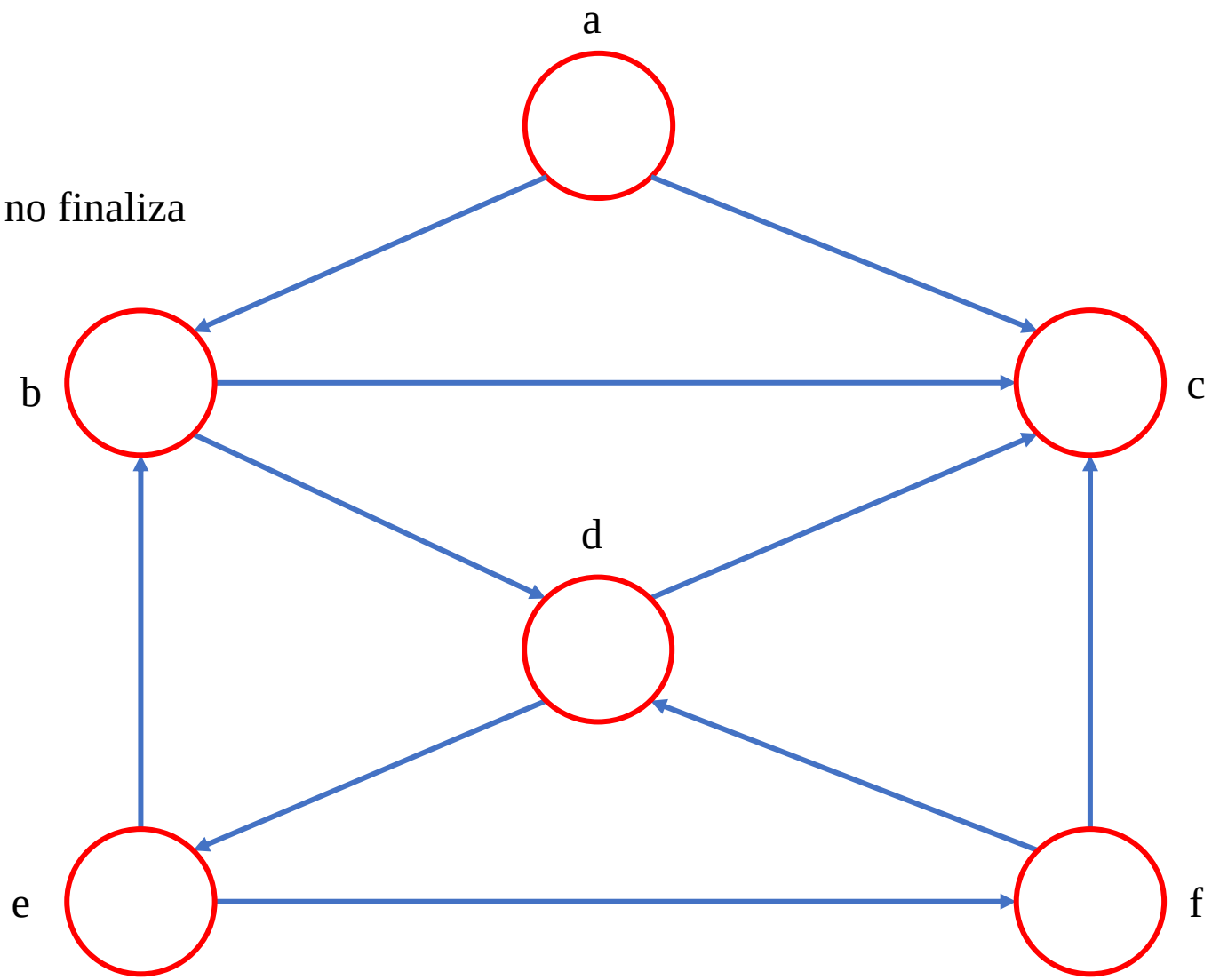


Vistos, todavía no finaliza



Finaliza



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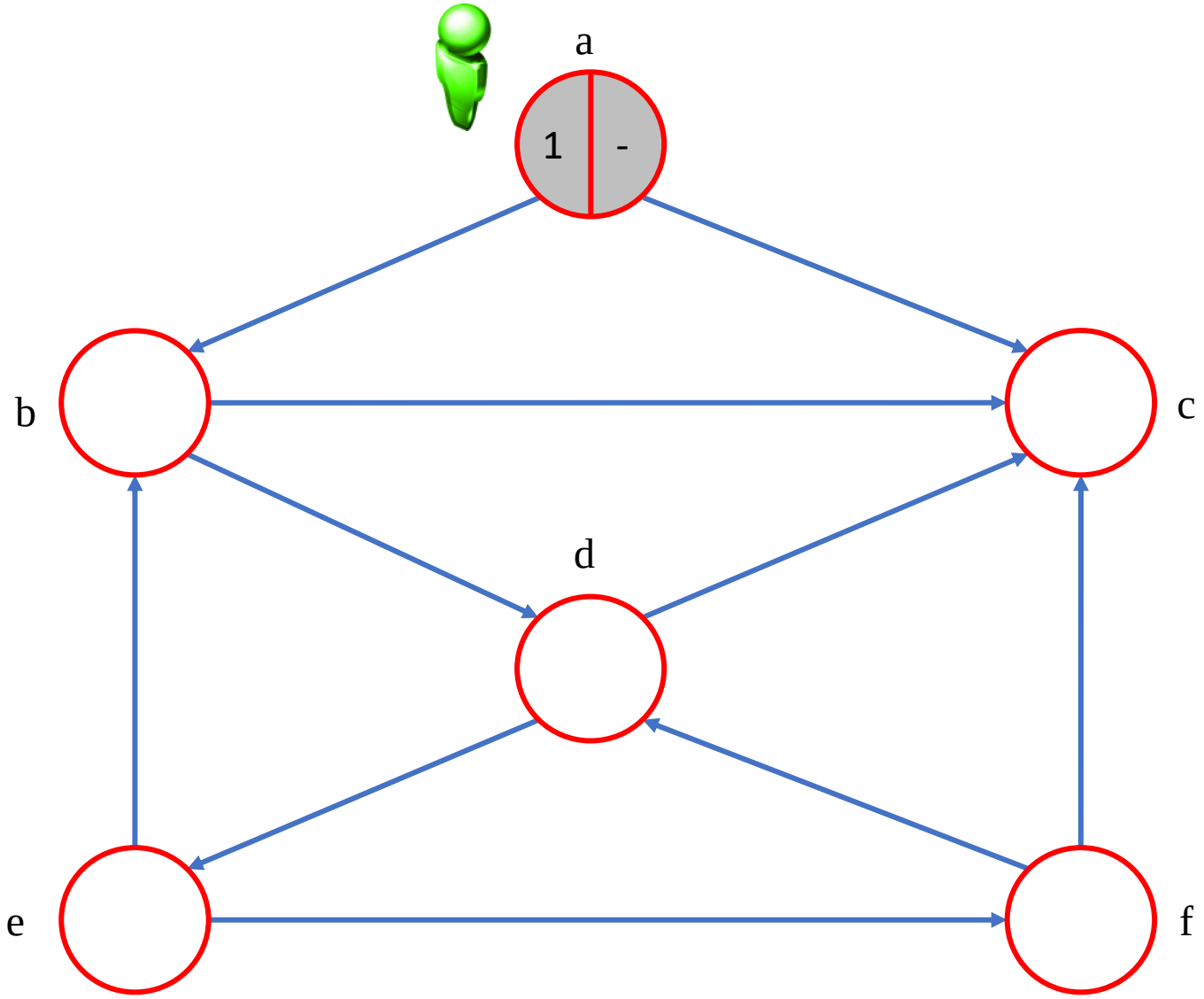

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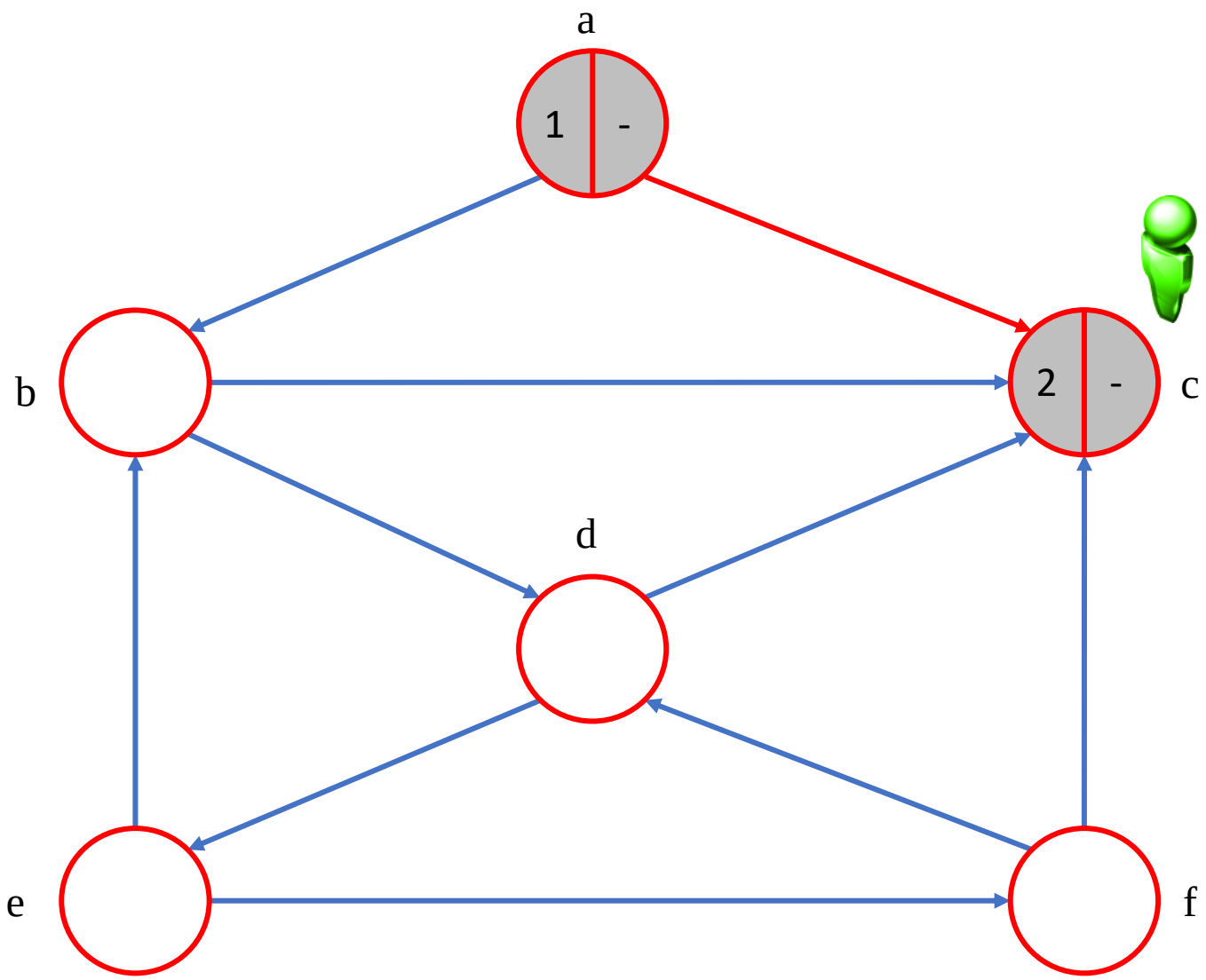

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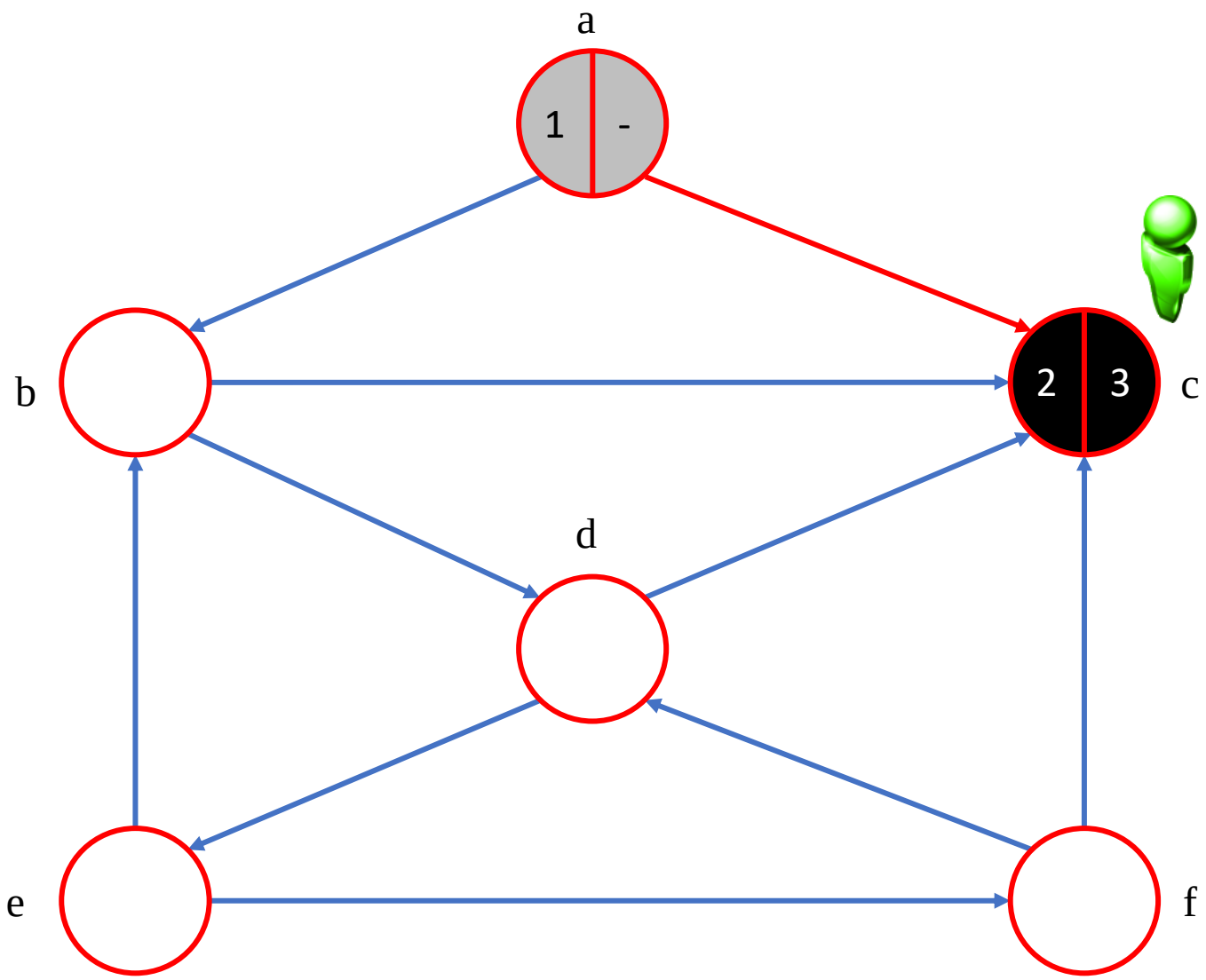

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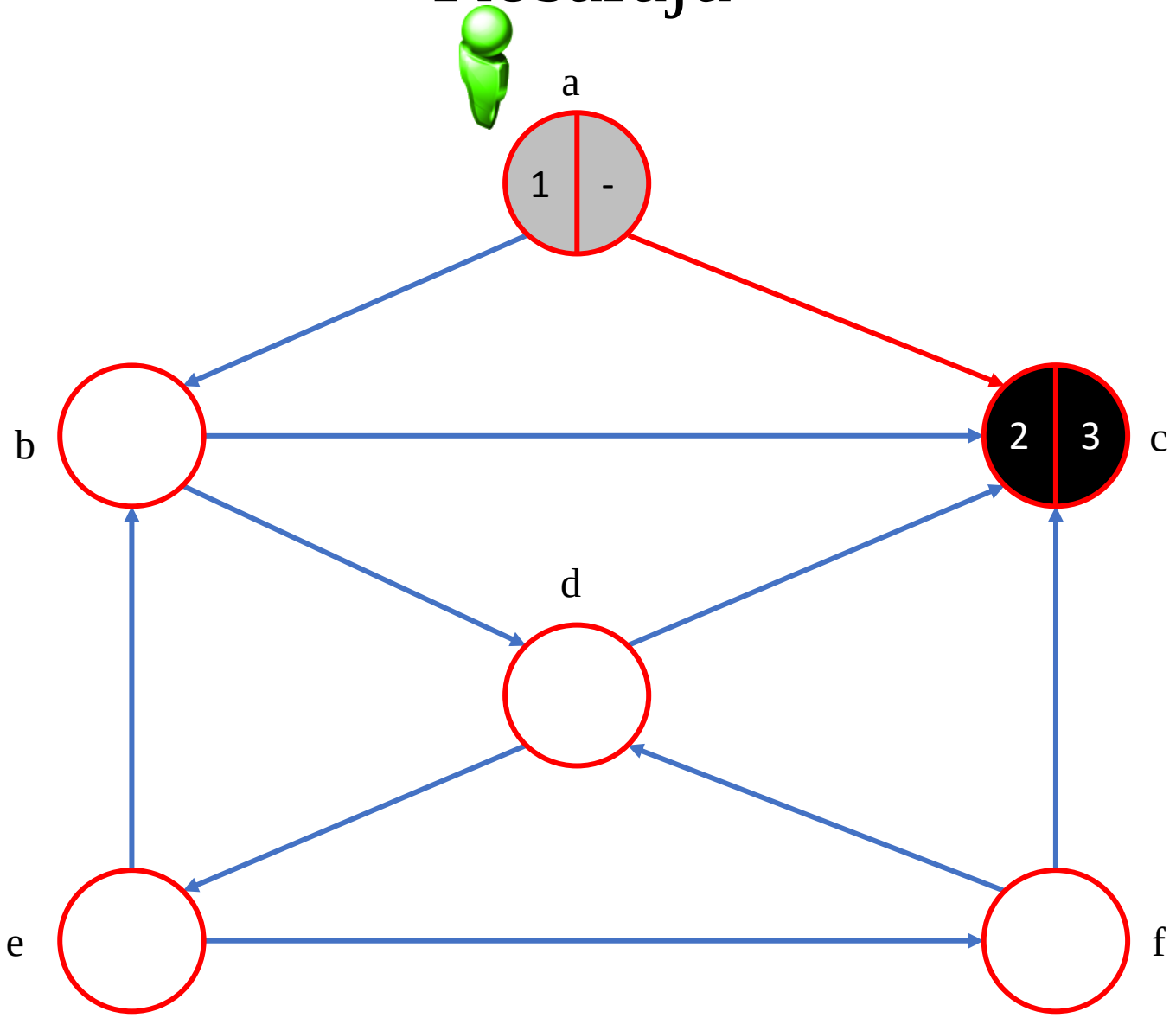

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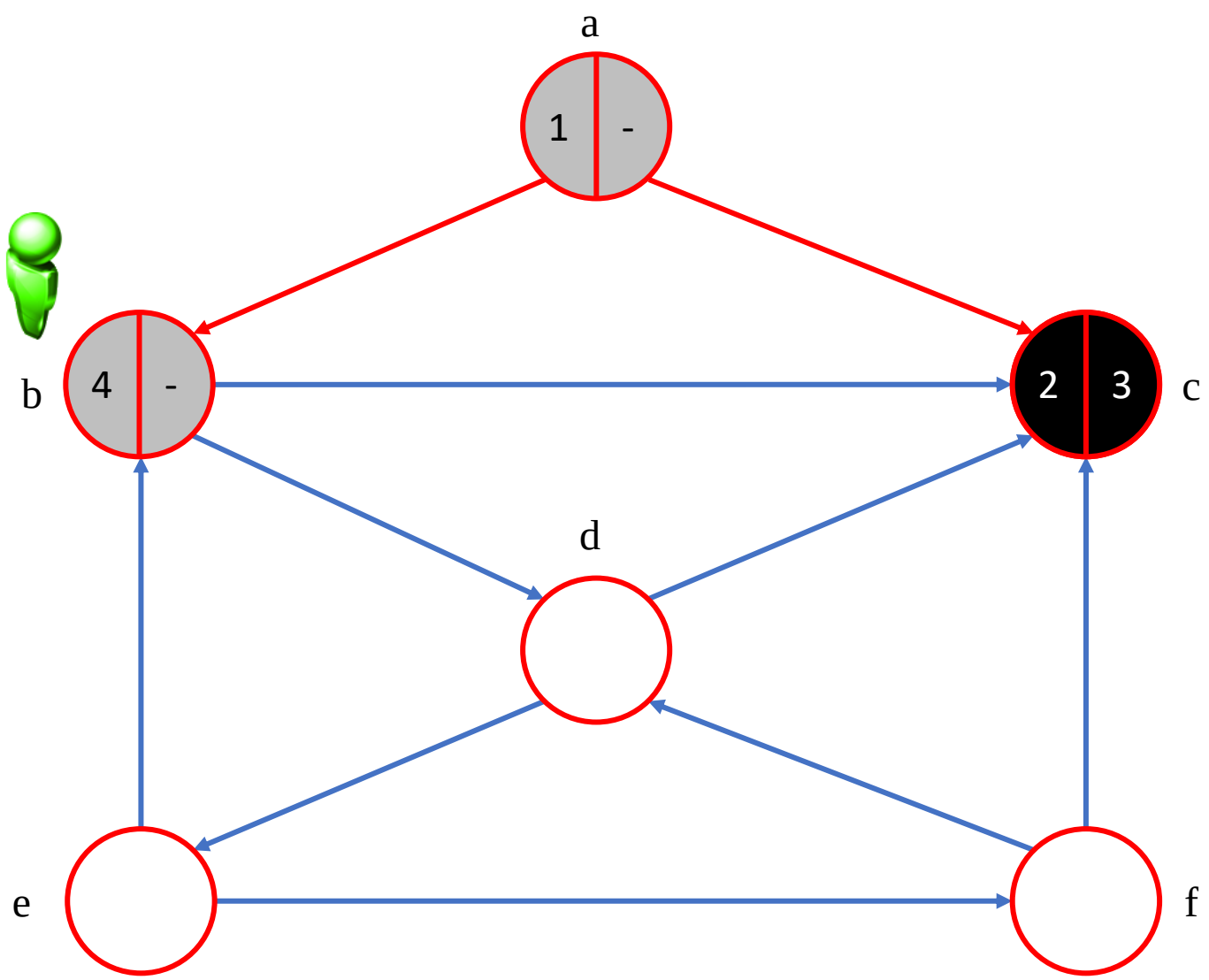

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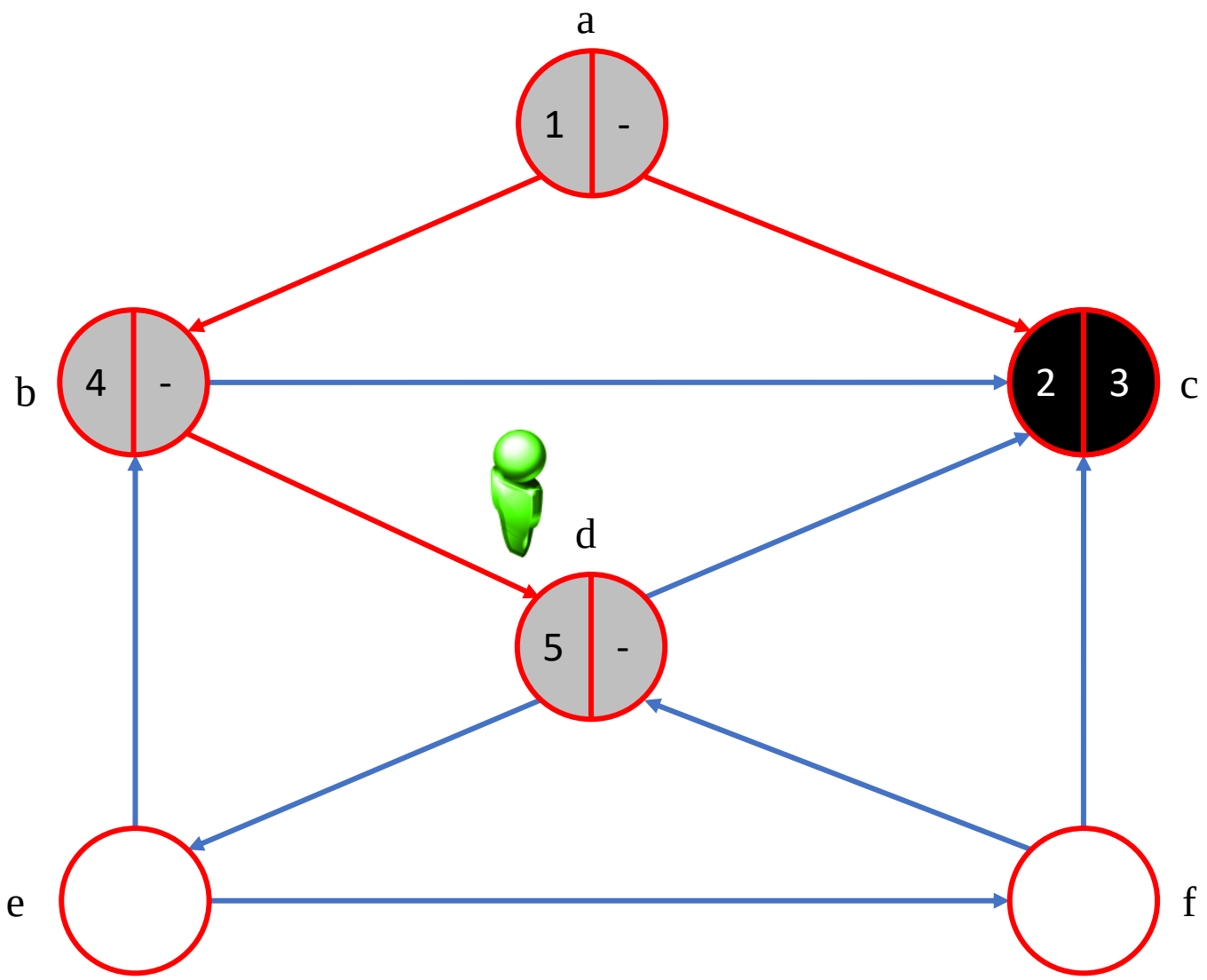

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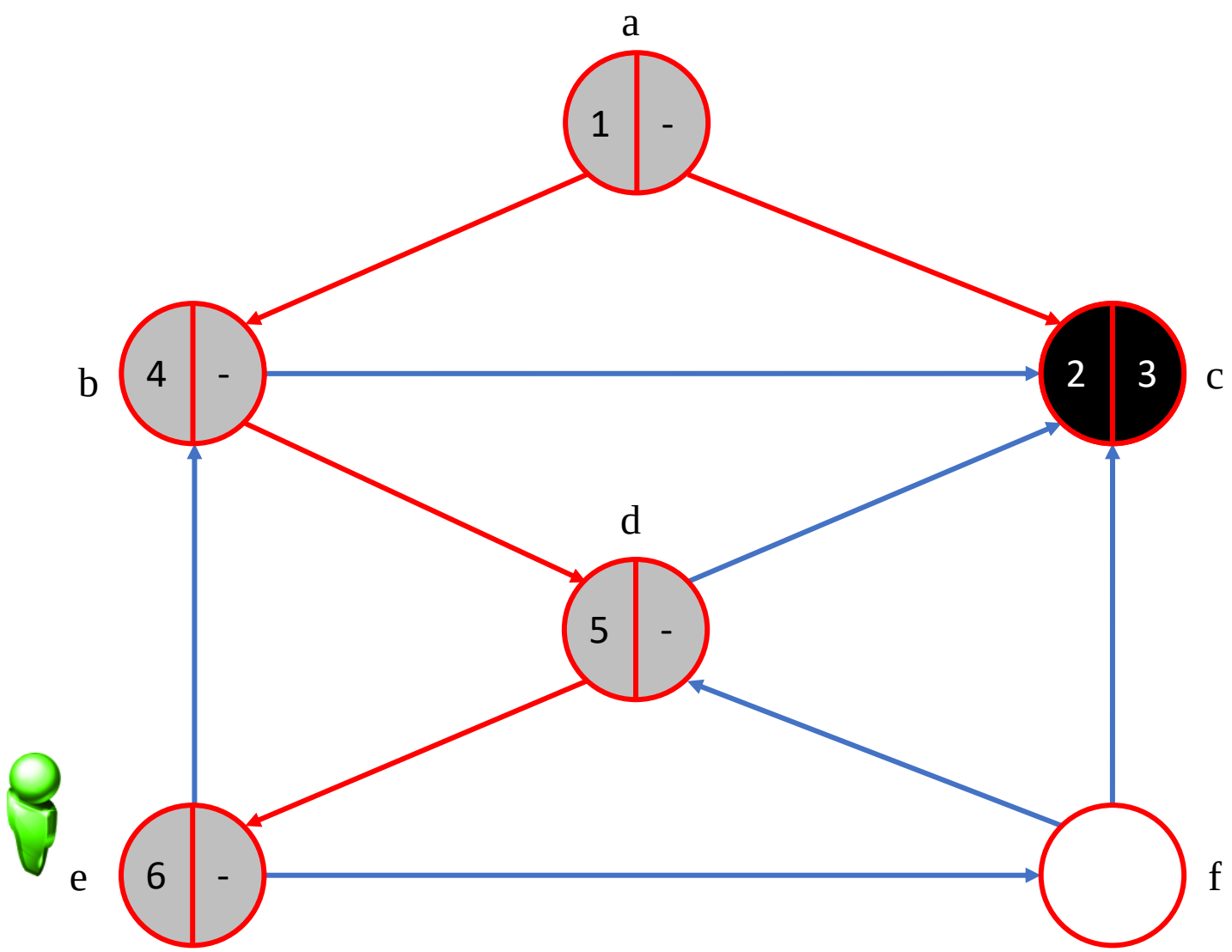

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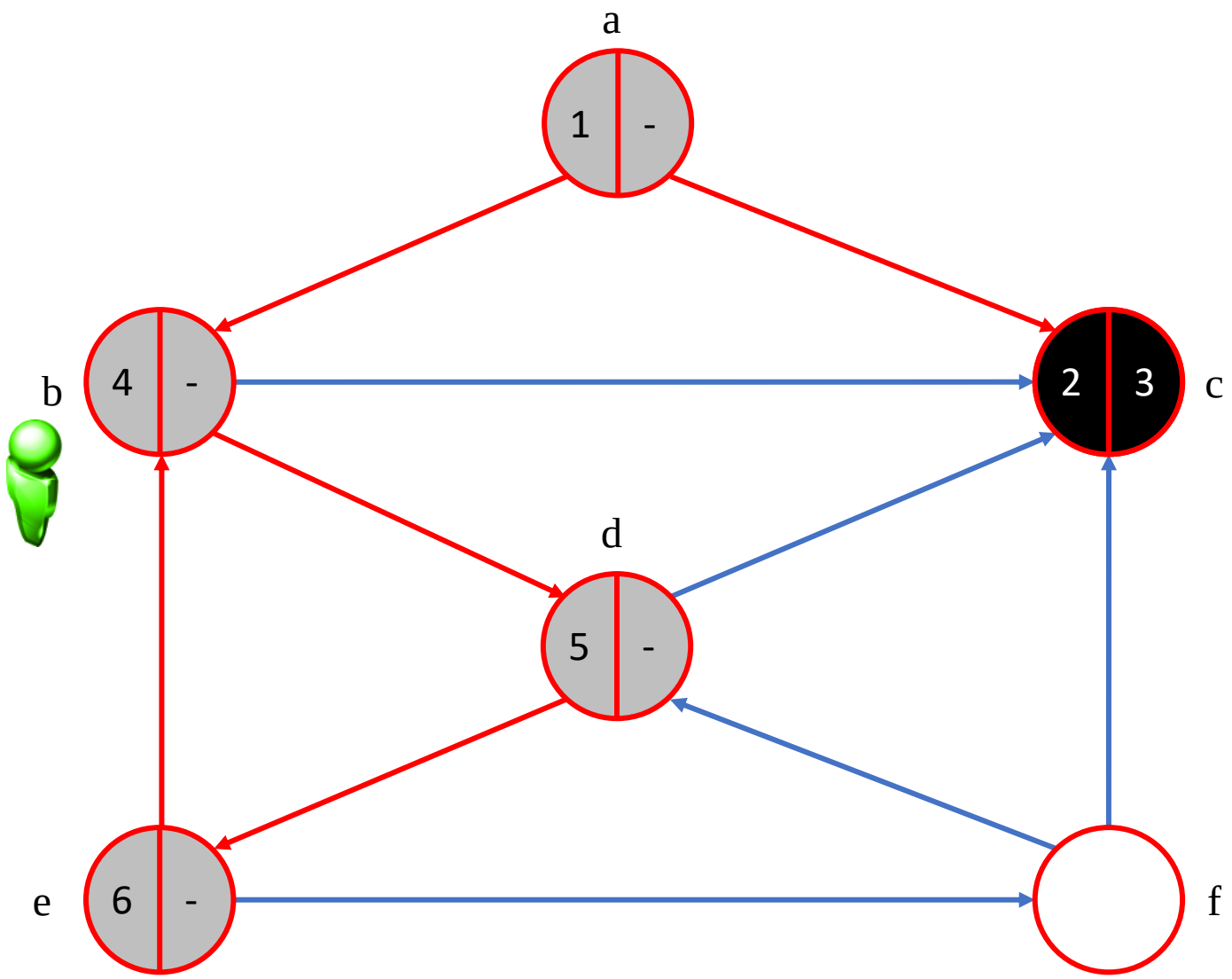

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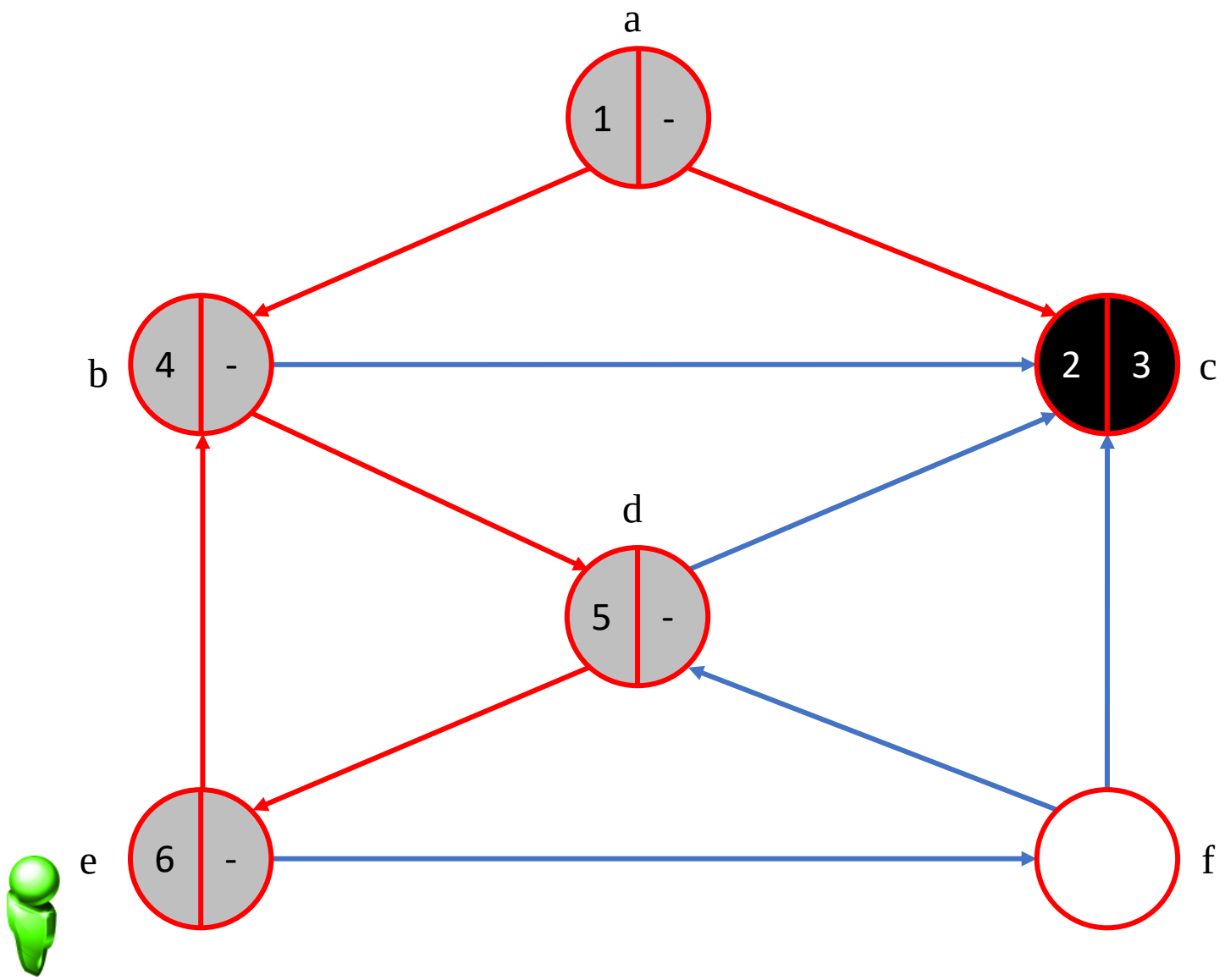

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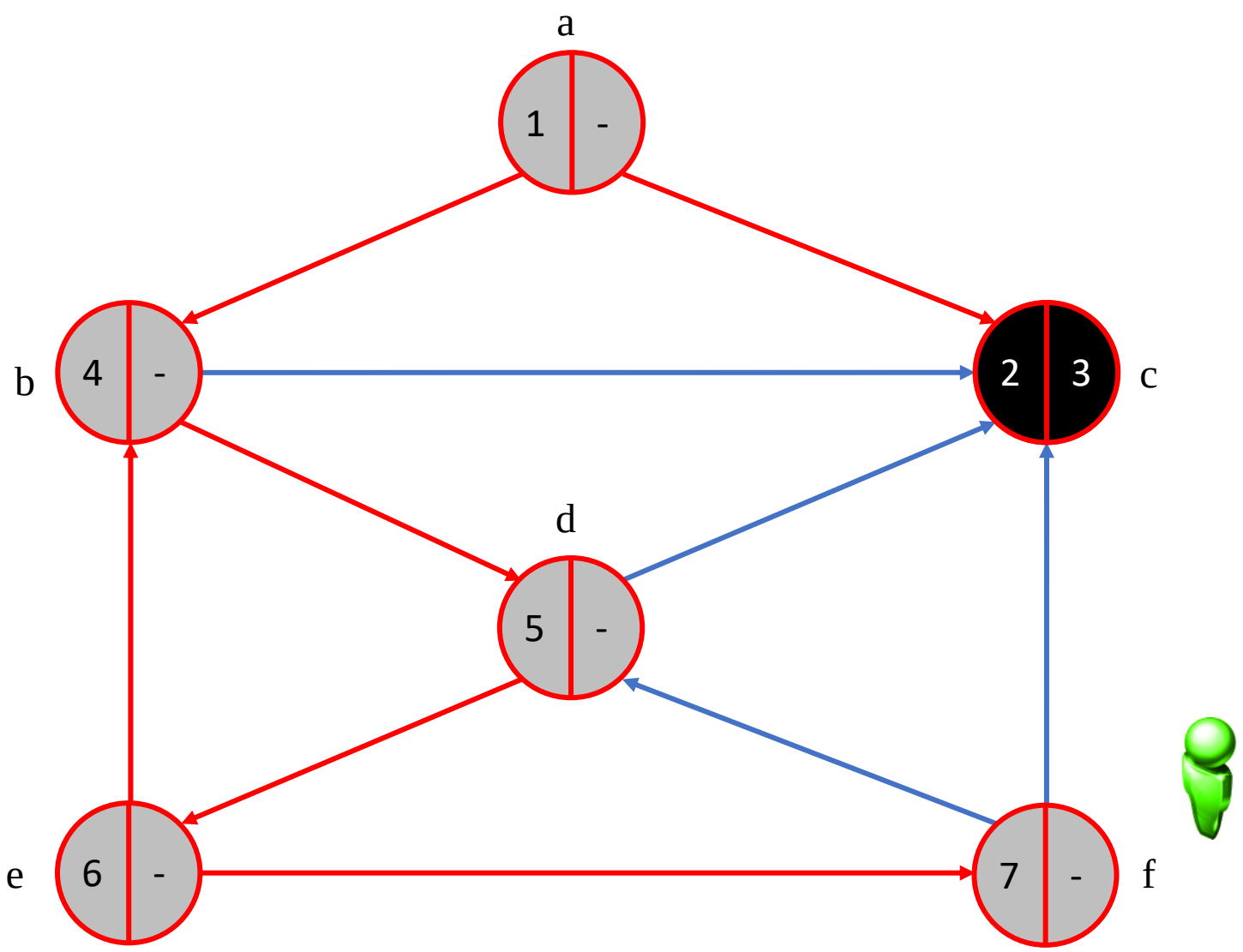

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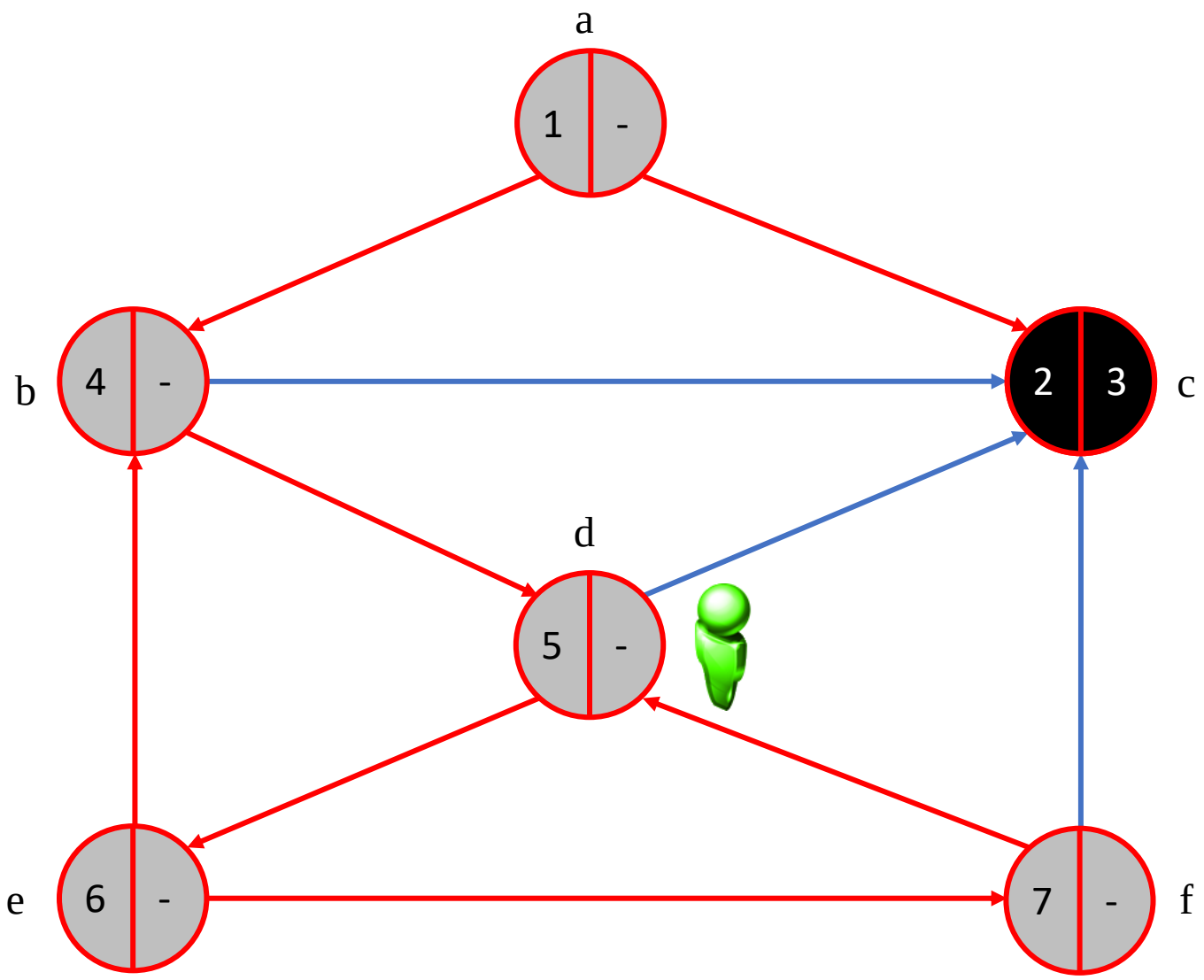

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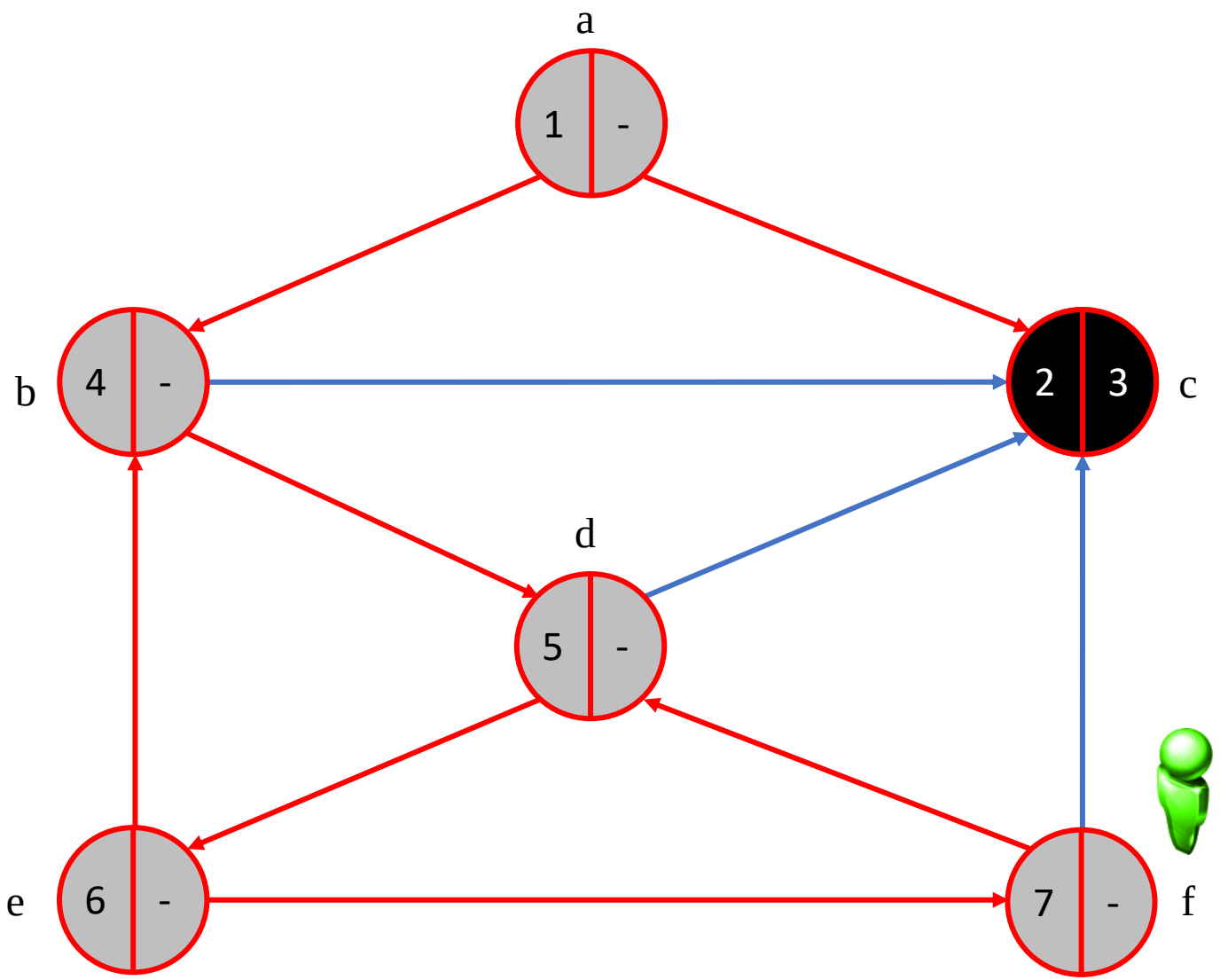

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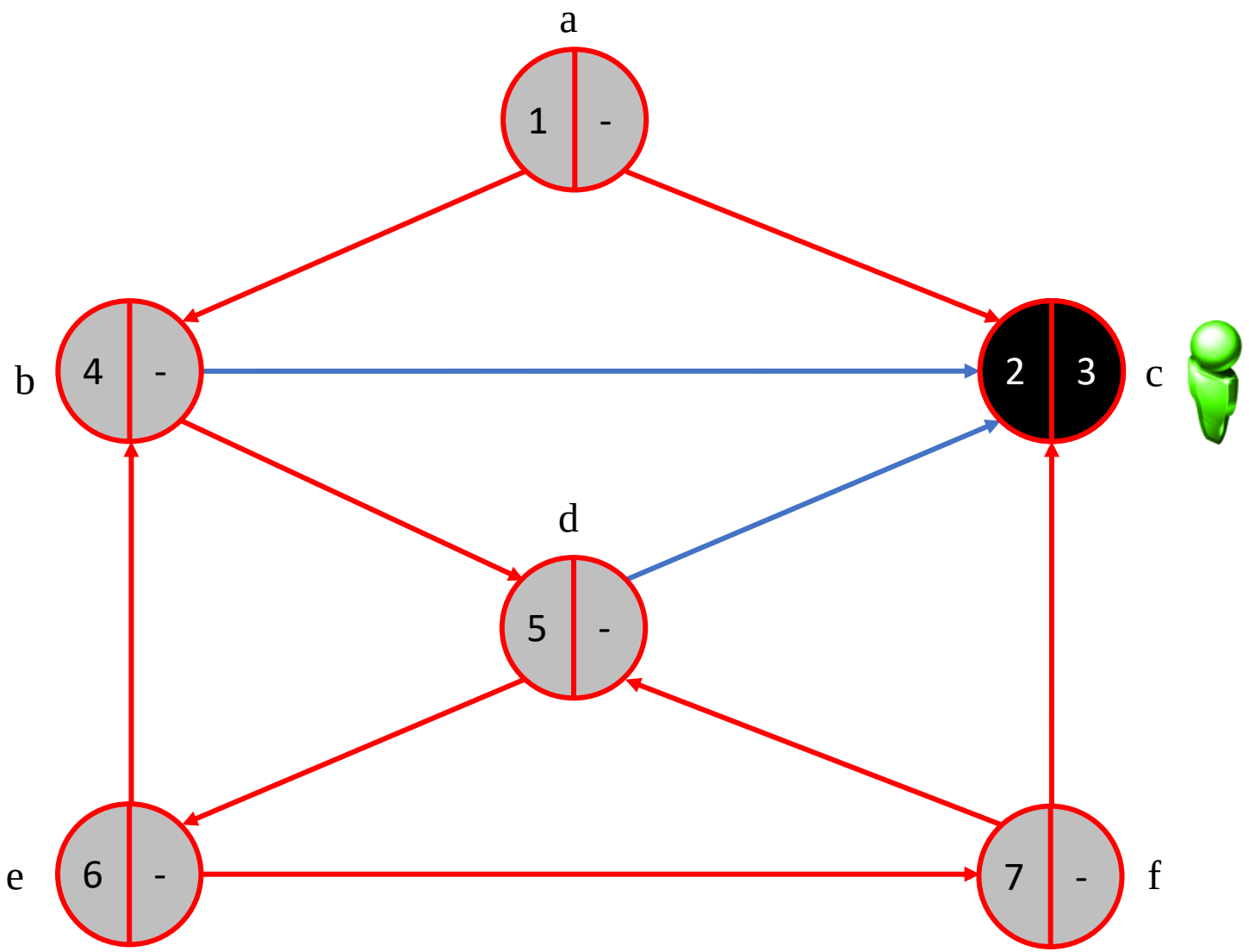

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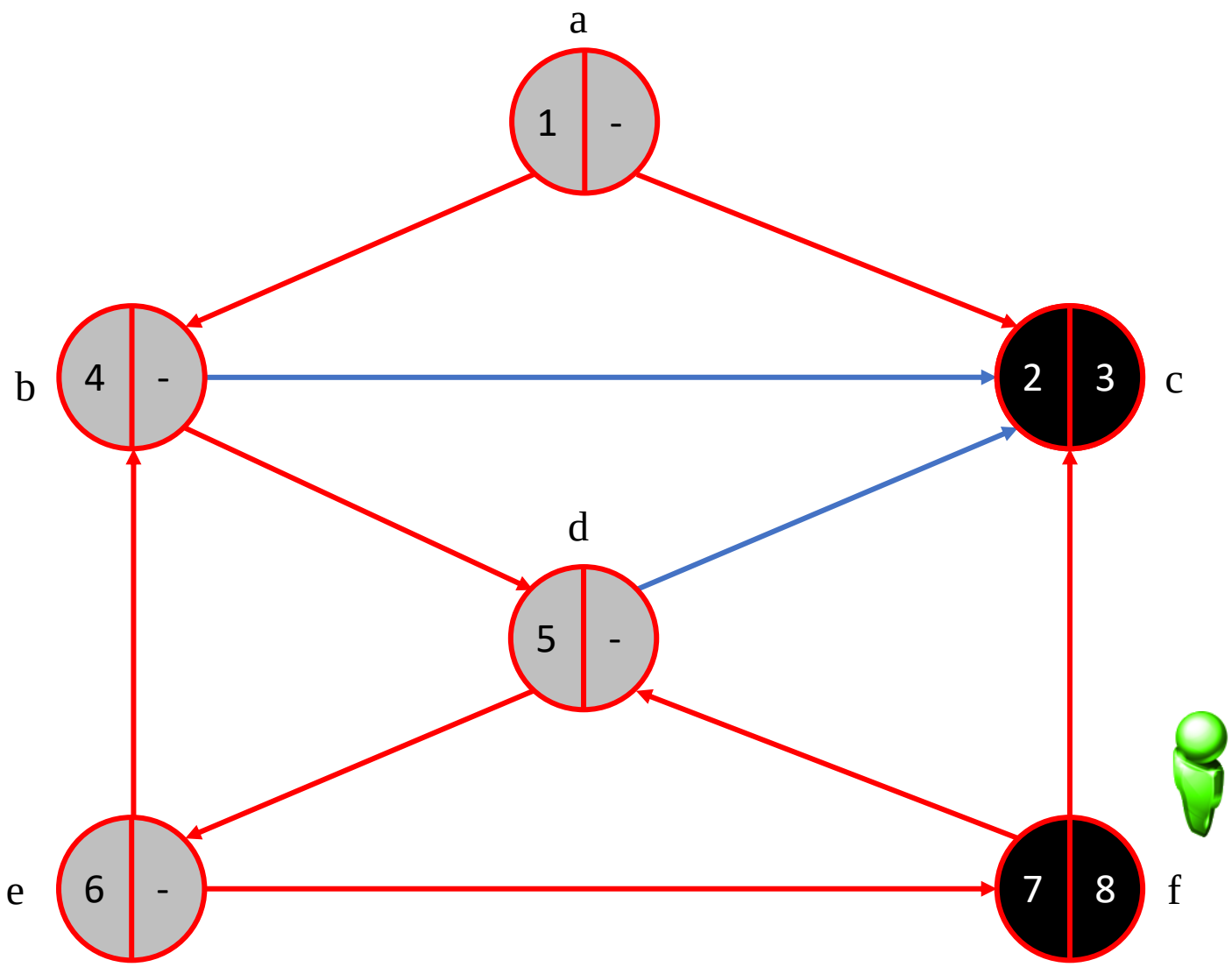

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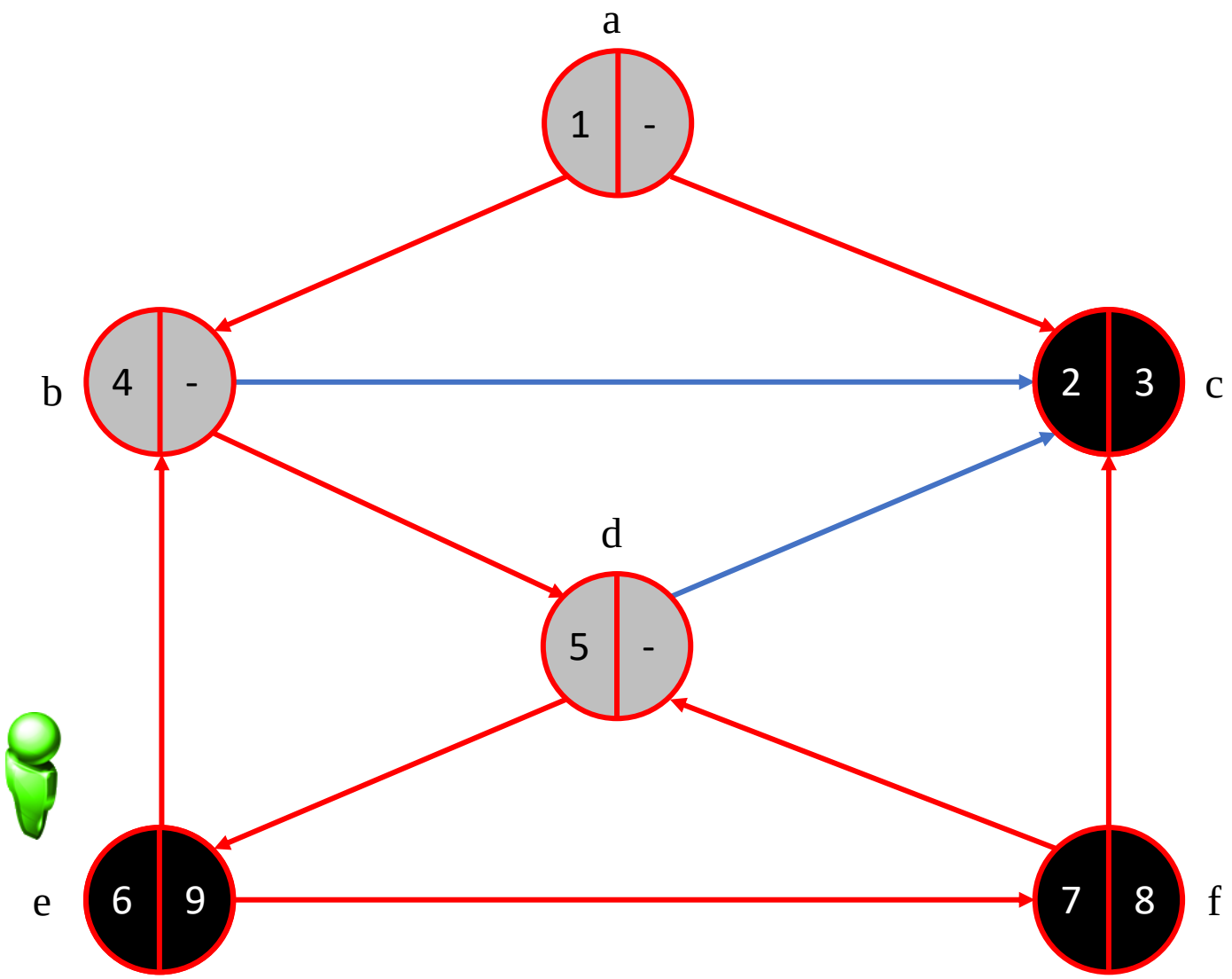

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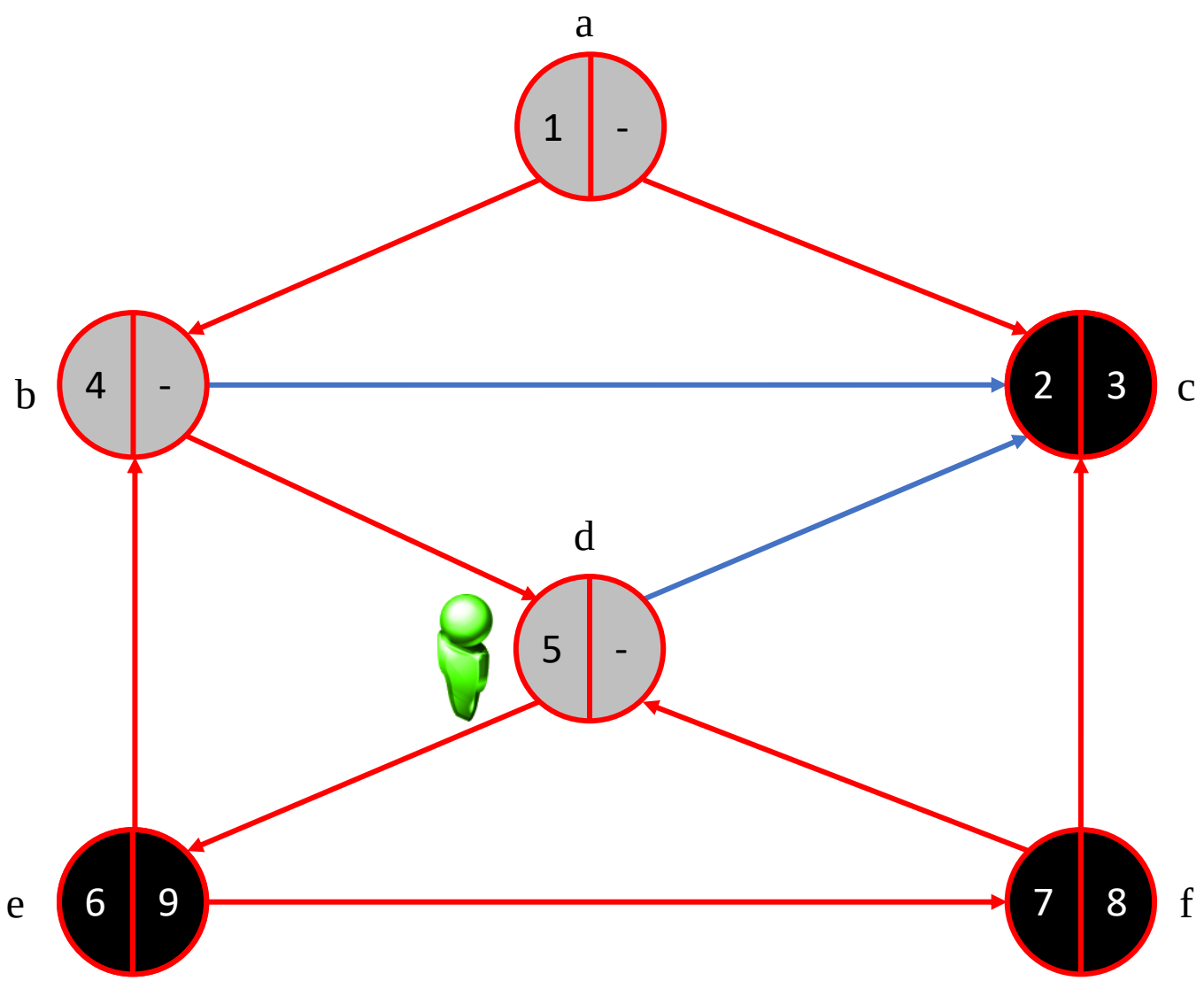

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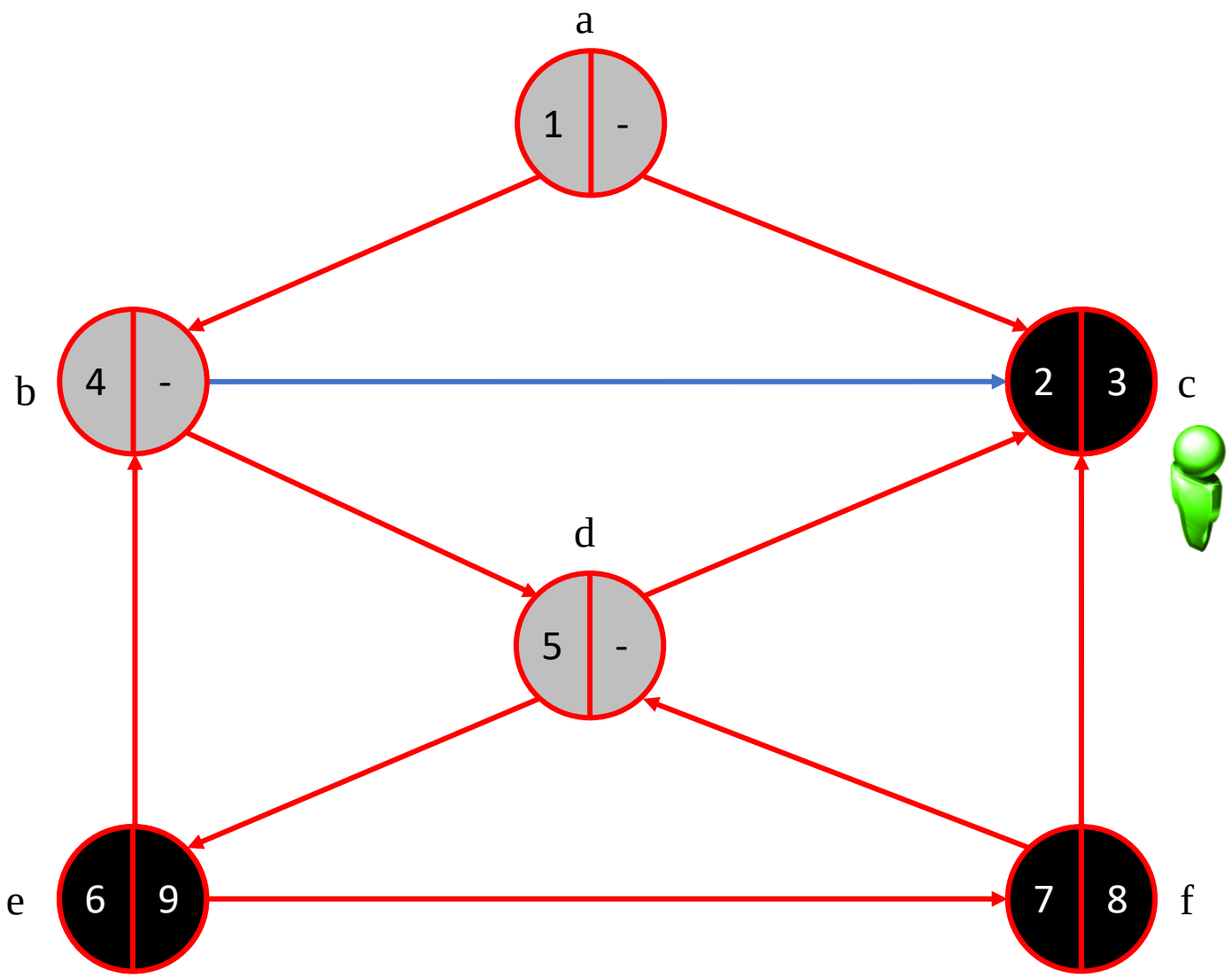

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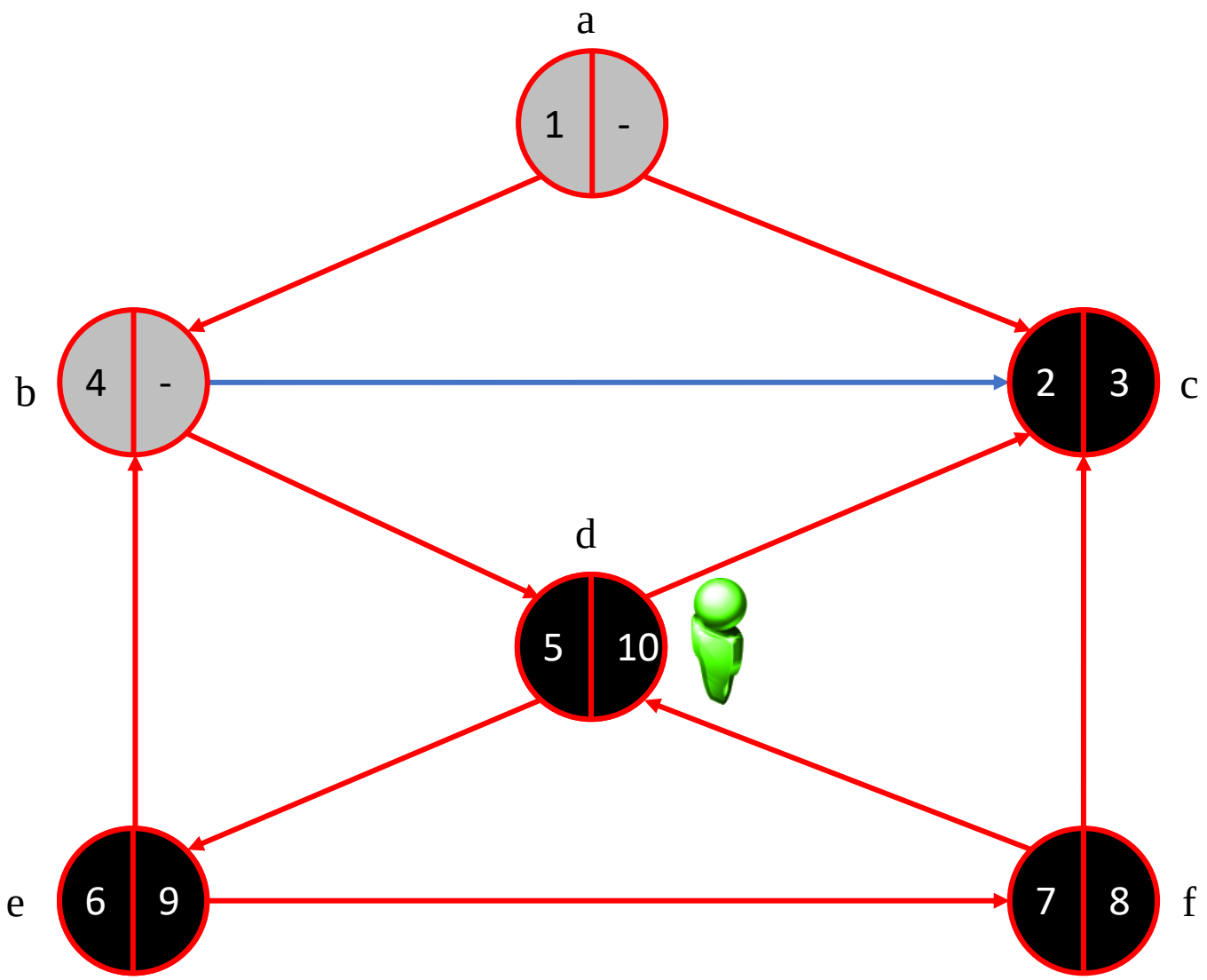

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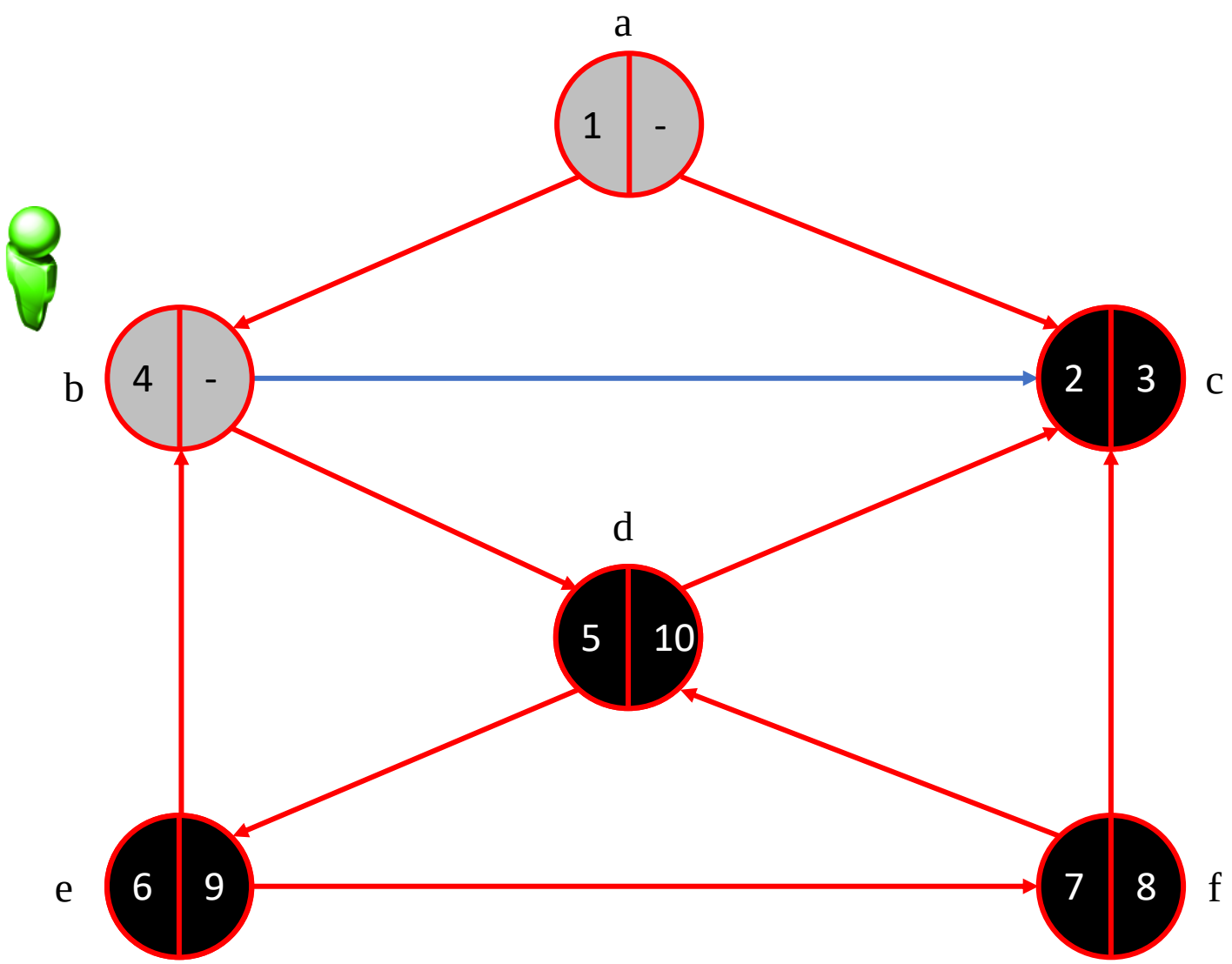

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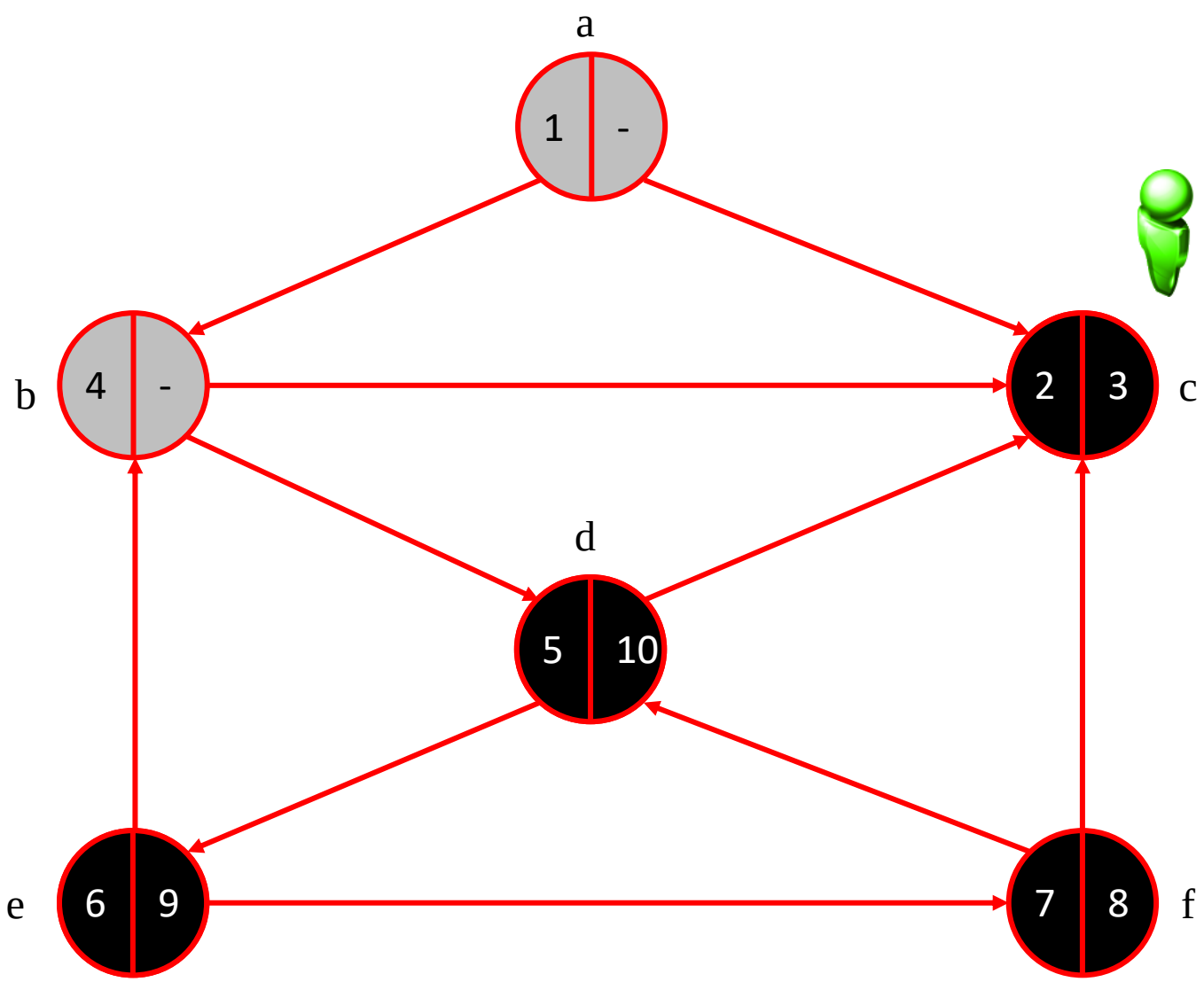

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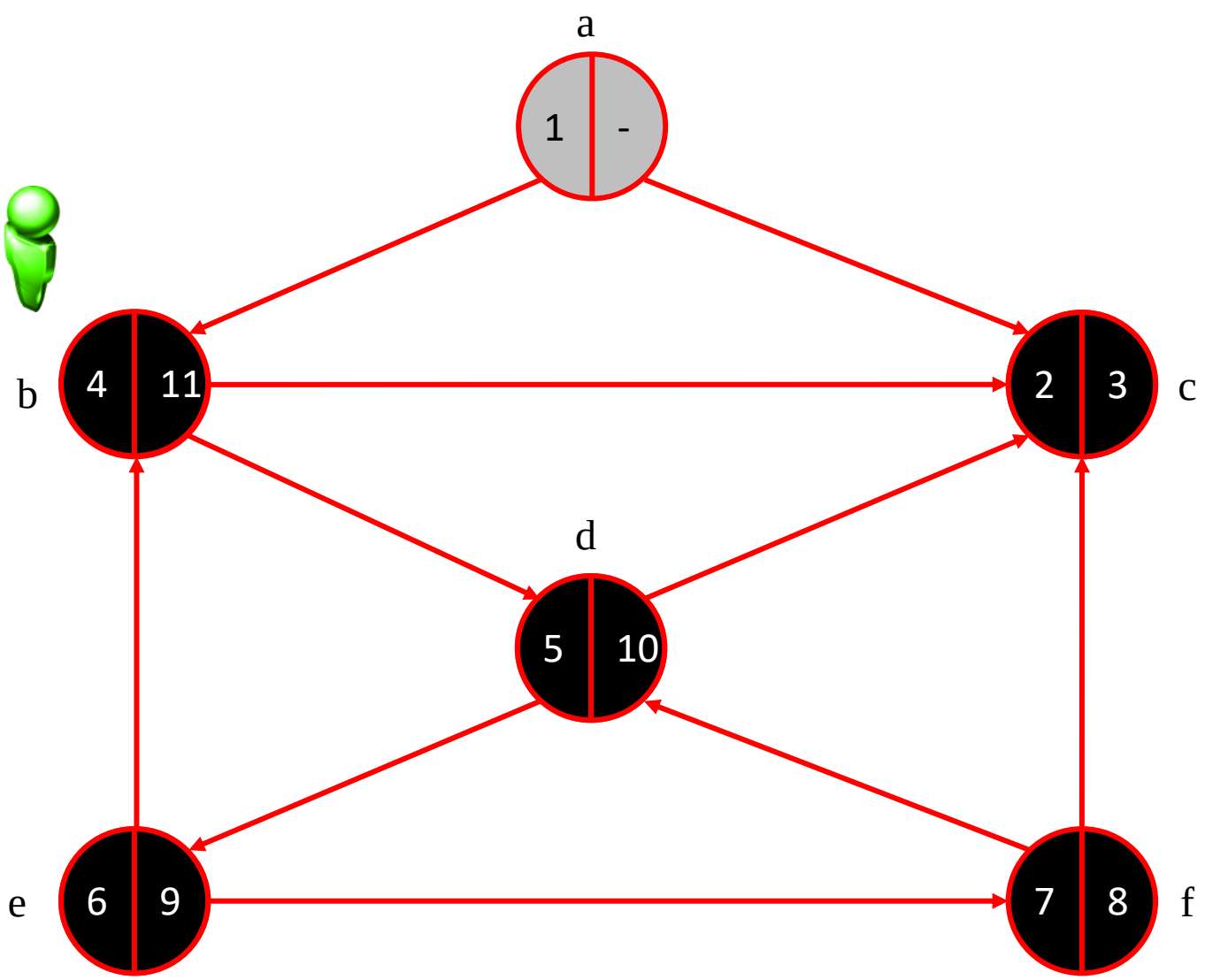

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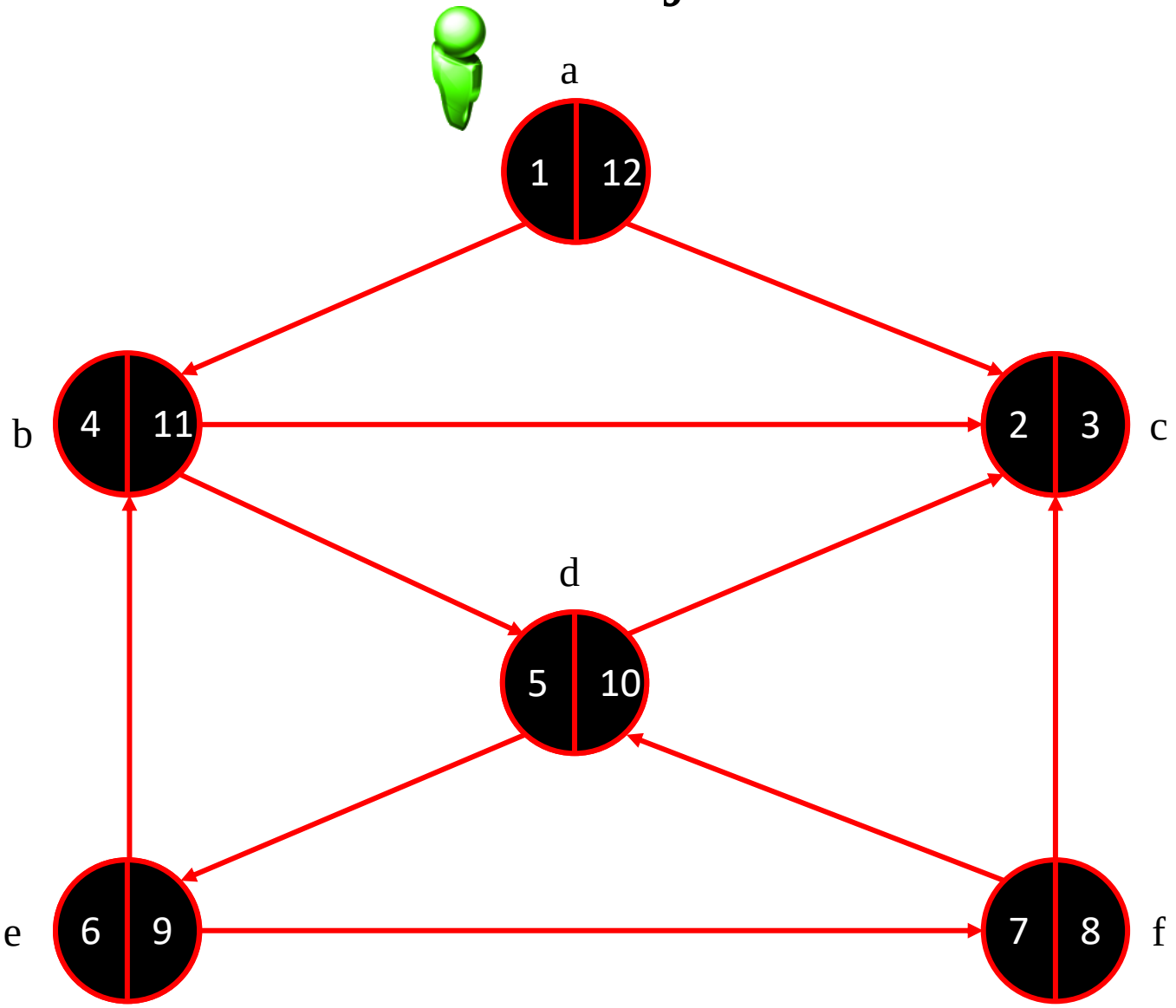

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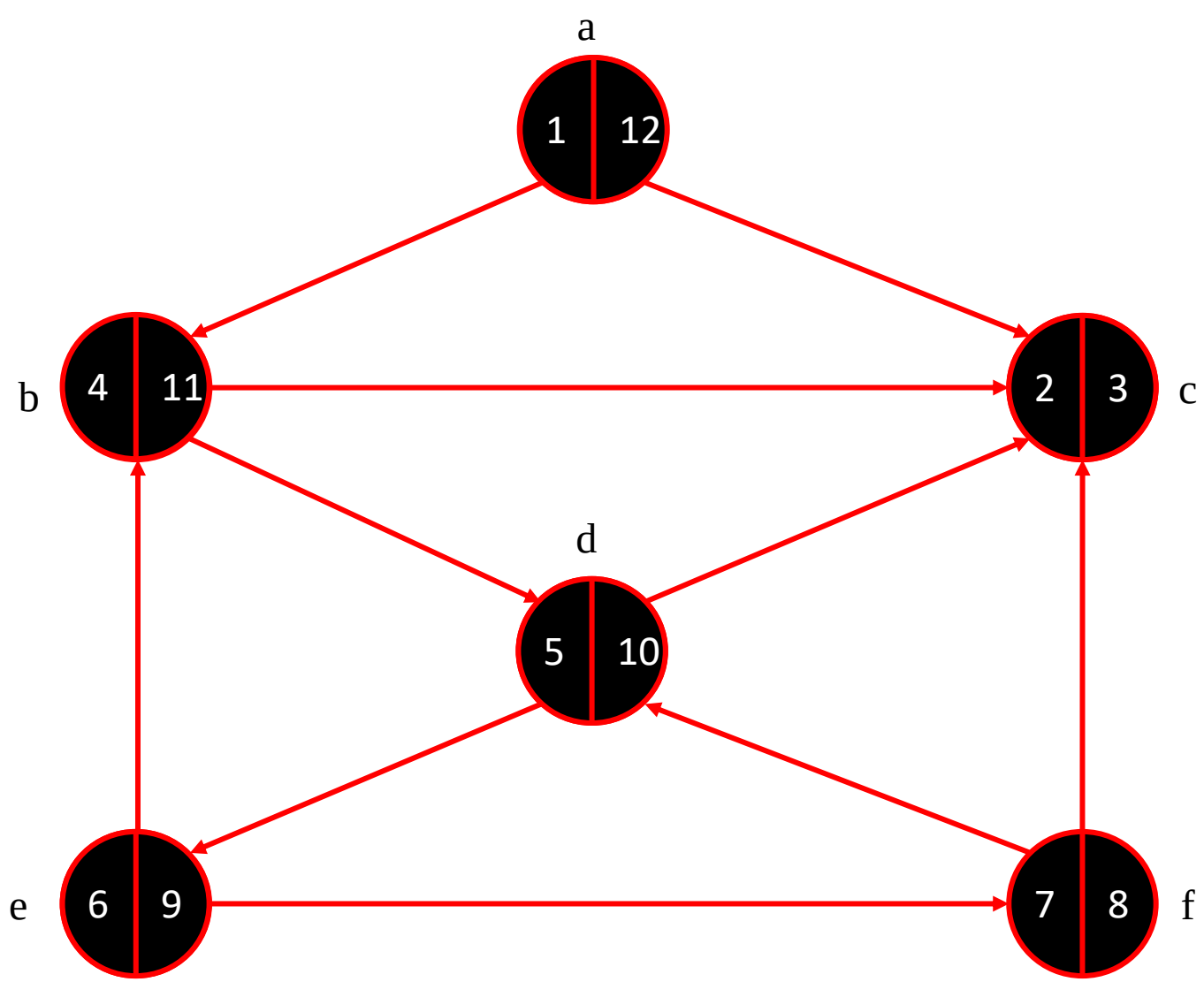

$\{c, f, e, d, b, a\}$

$\{a, b, d, e, f, c\}$

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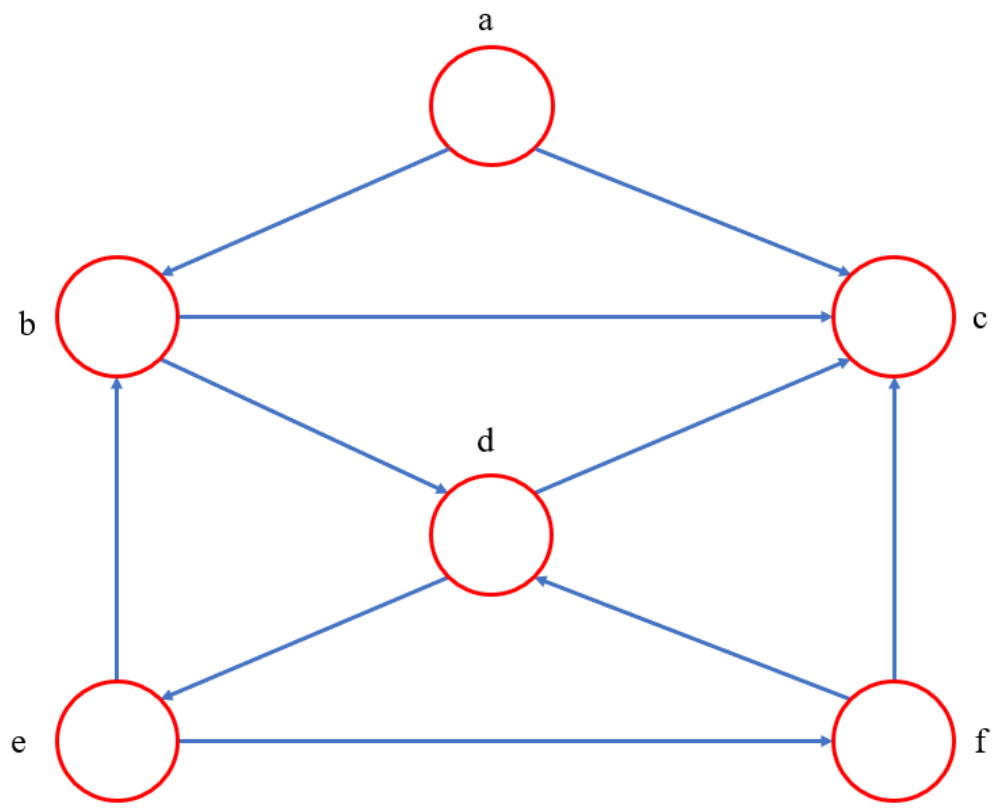

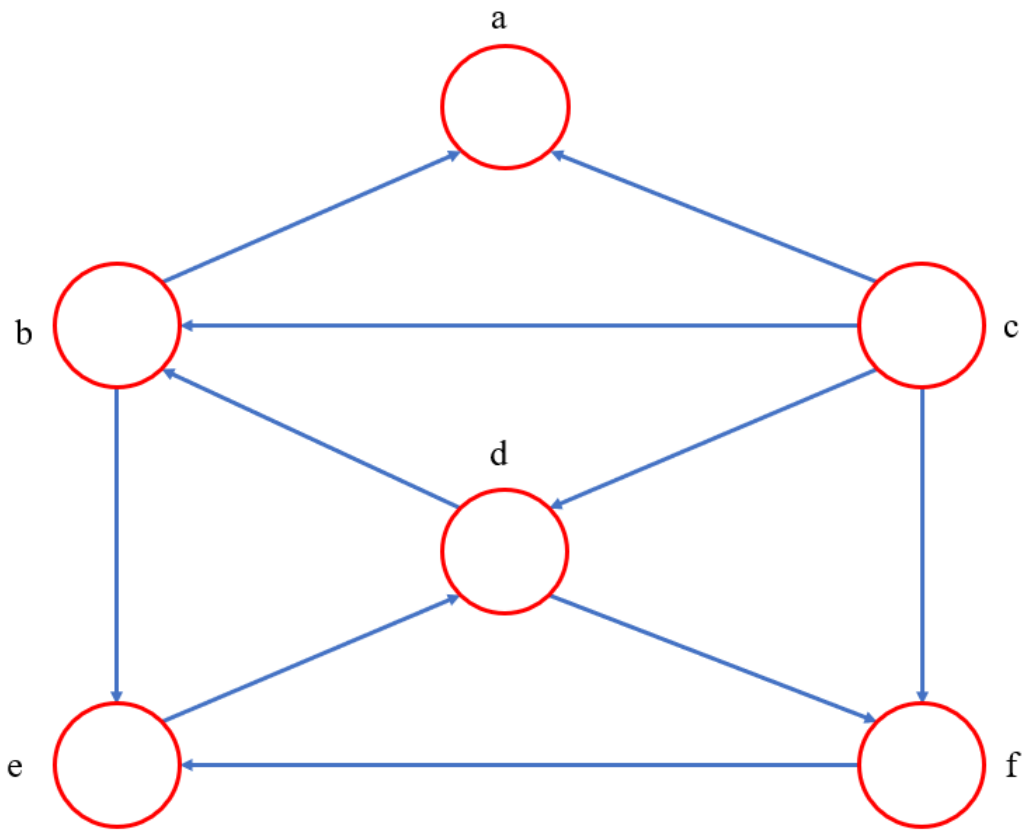

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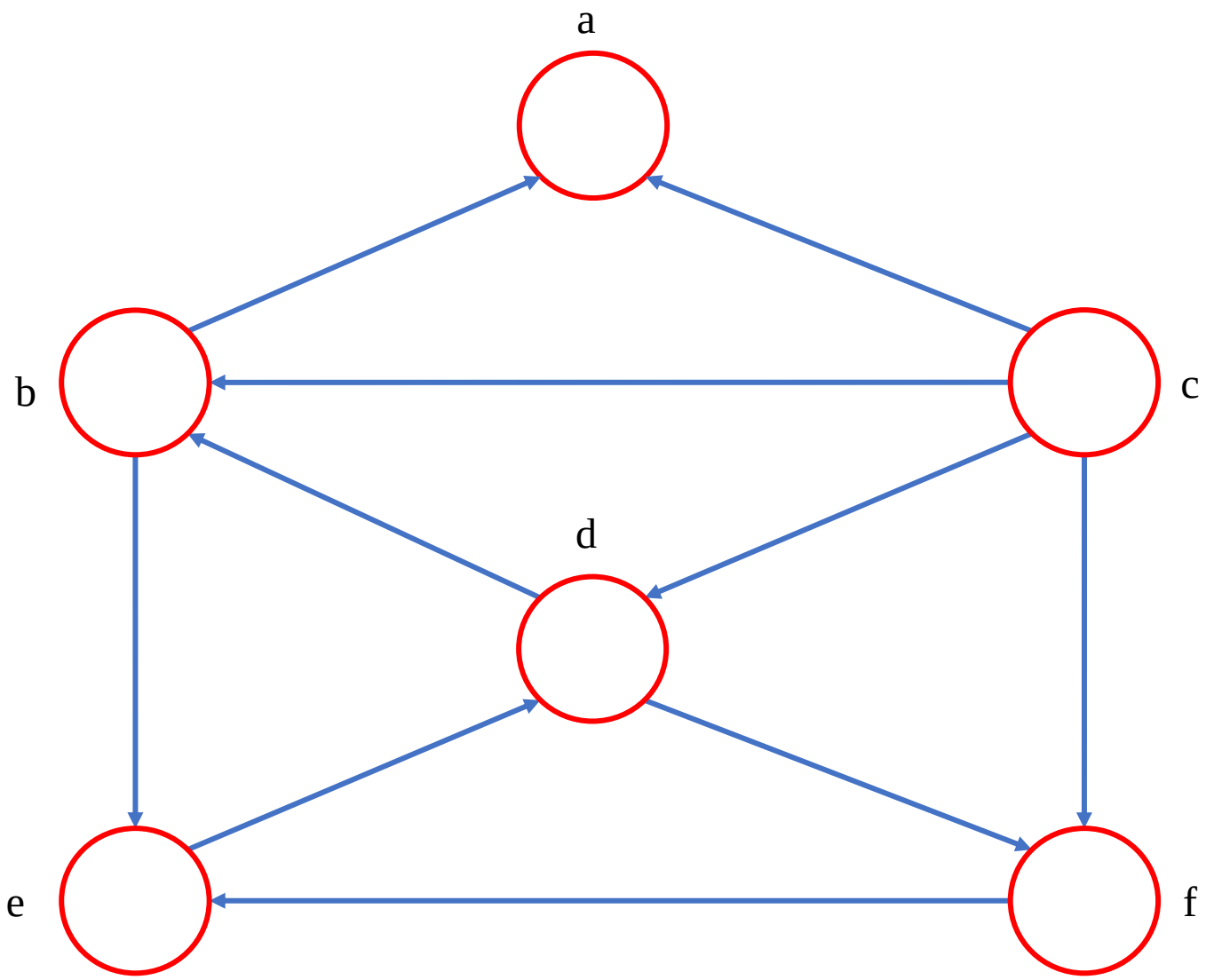
$\{a, b, d, e, f, c\}$



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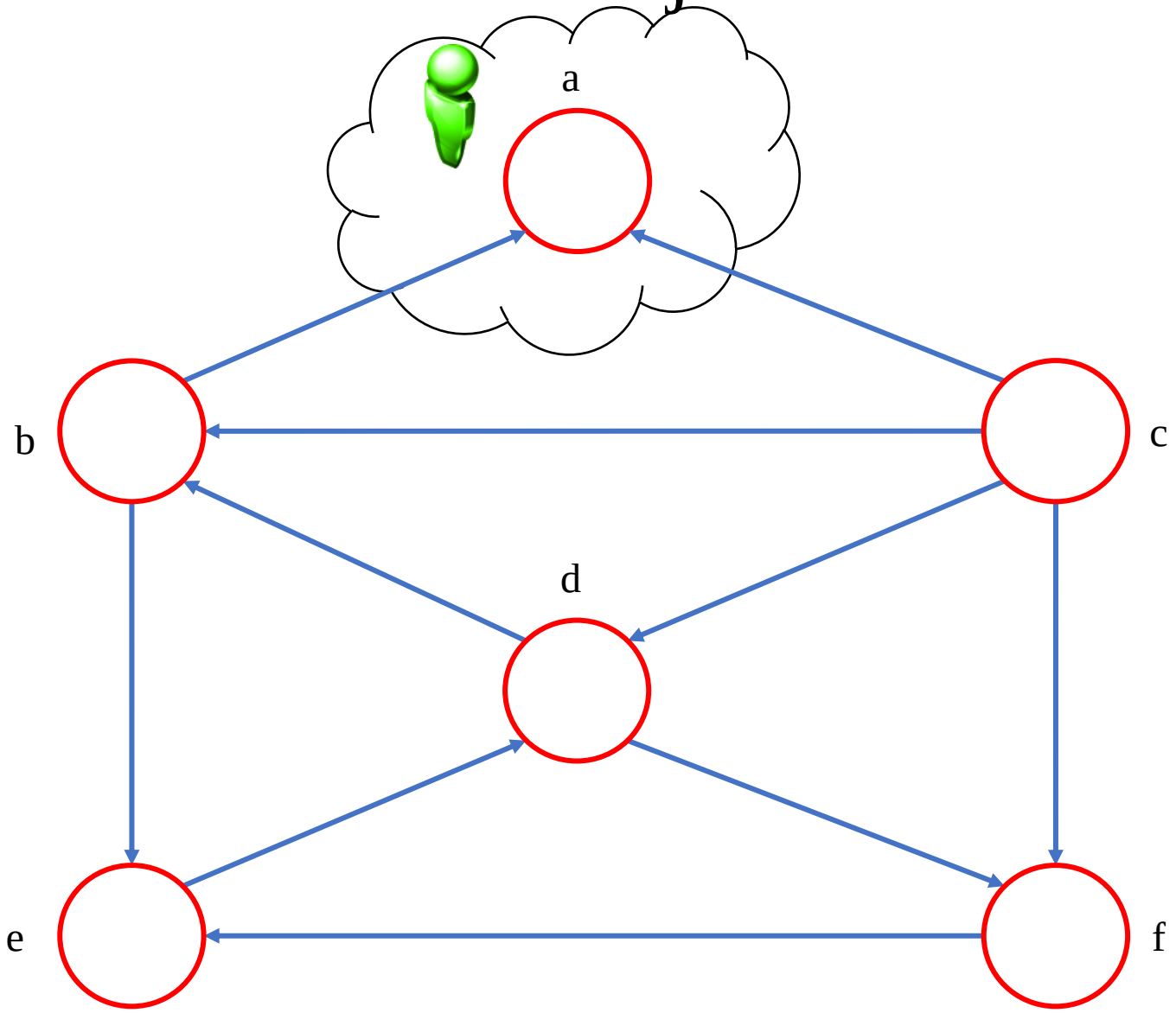
SCCs

$\{a\}$

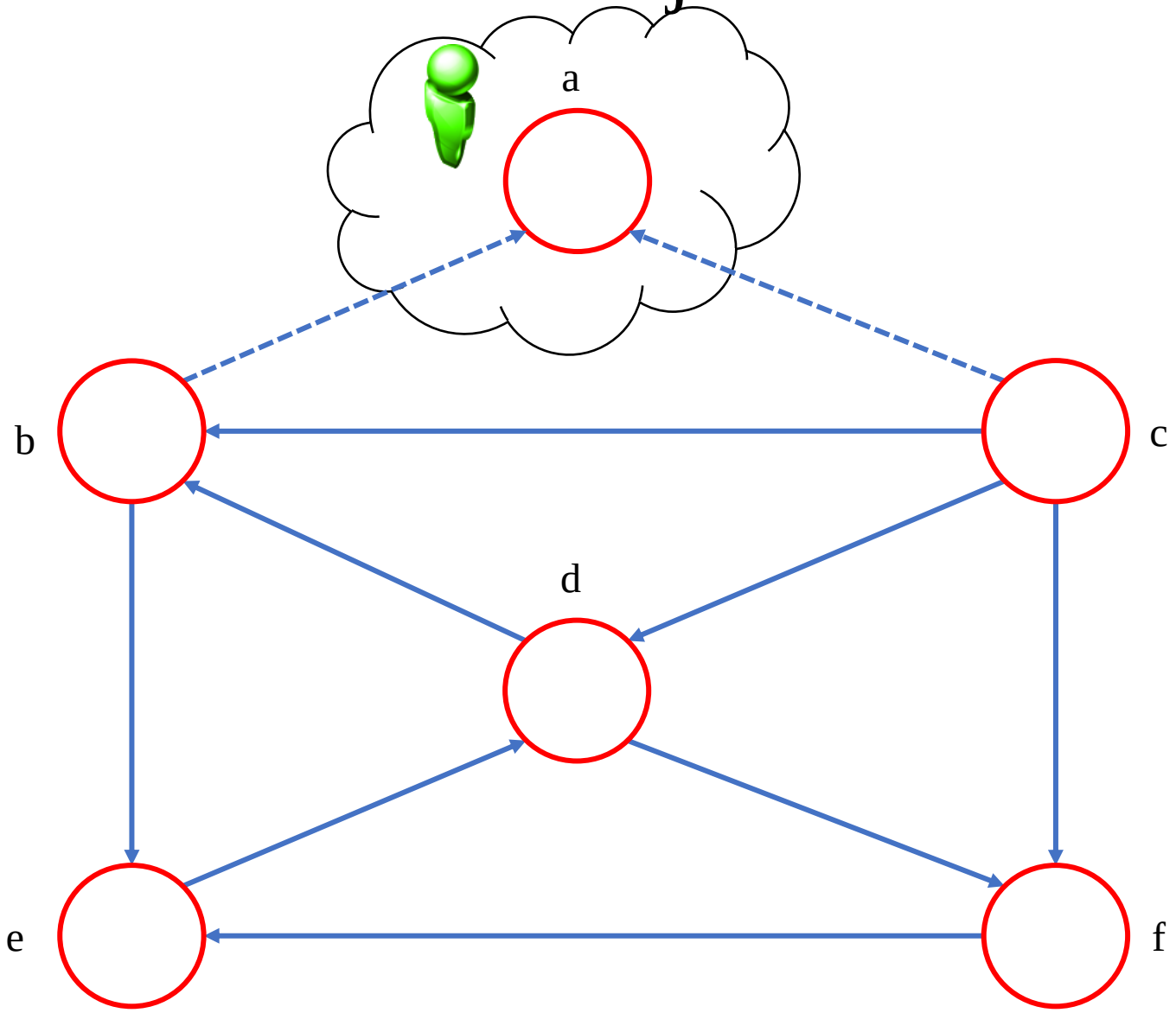


$\{a, b, d, e, f, c\}$

Kosaraju

SCCs

$\{a\}$



$\{a, b, d, e, f, c\}$

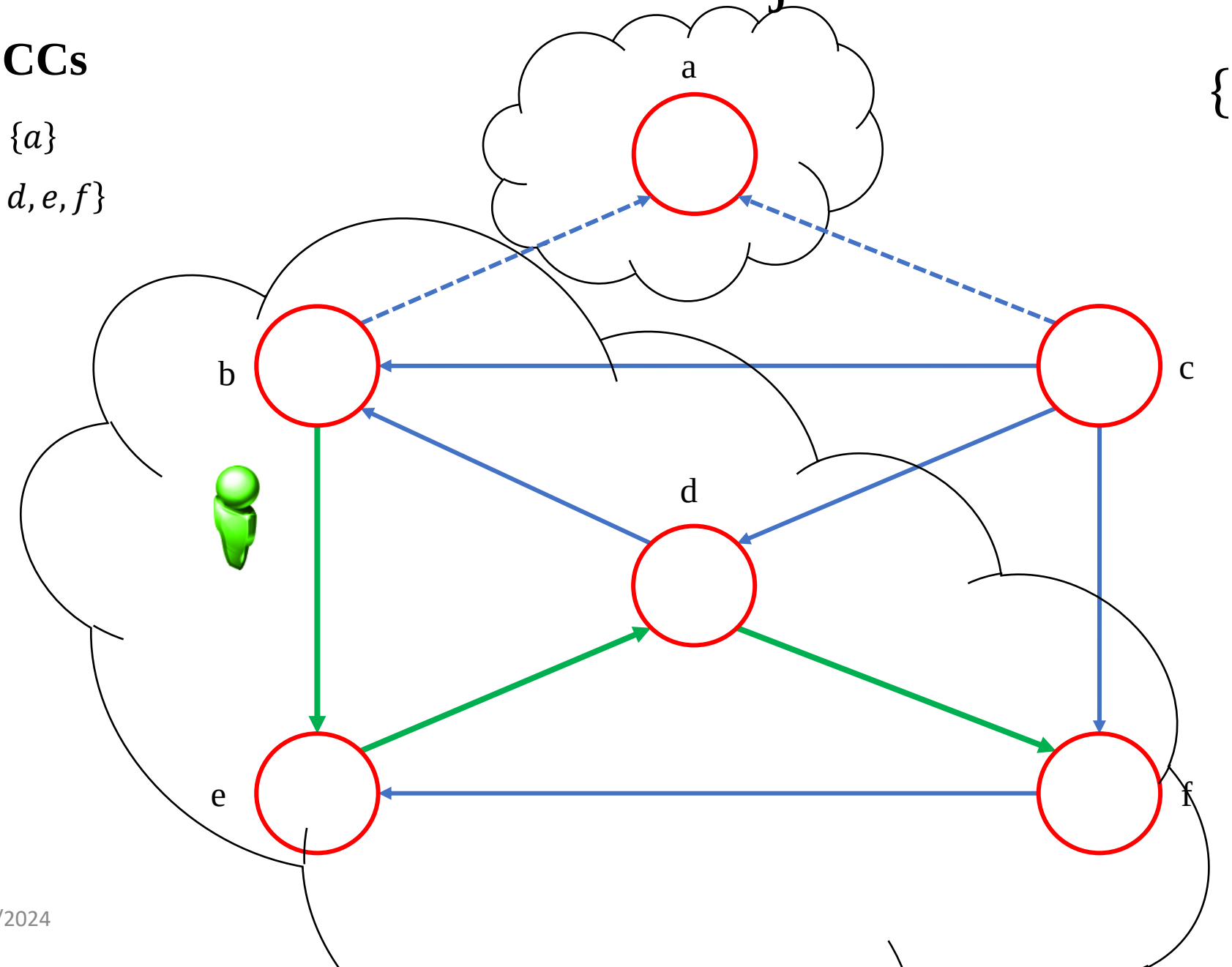
Kosaraju

SCCs

$\{a\}$

$\{b, d, e, f\}$

$\{a, b, d, e, f, c\}$



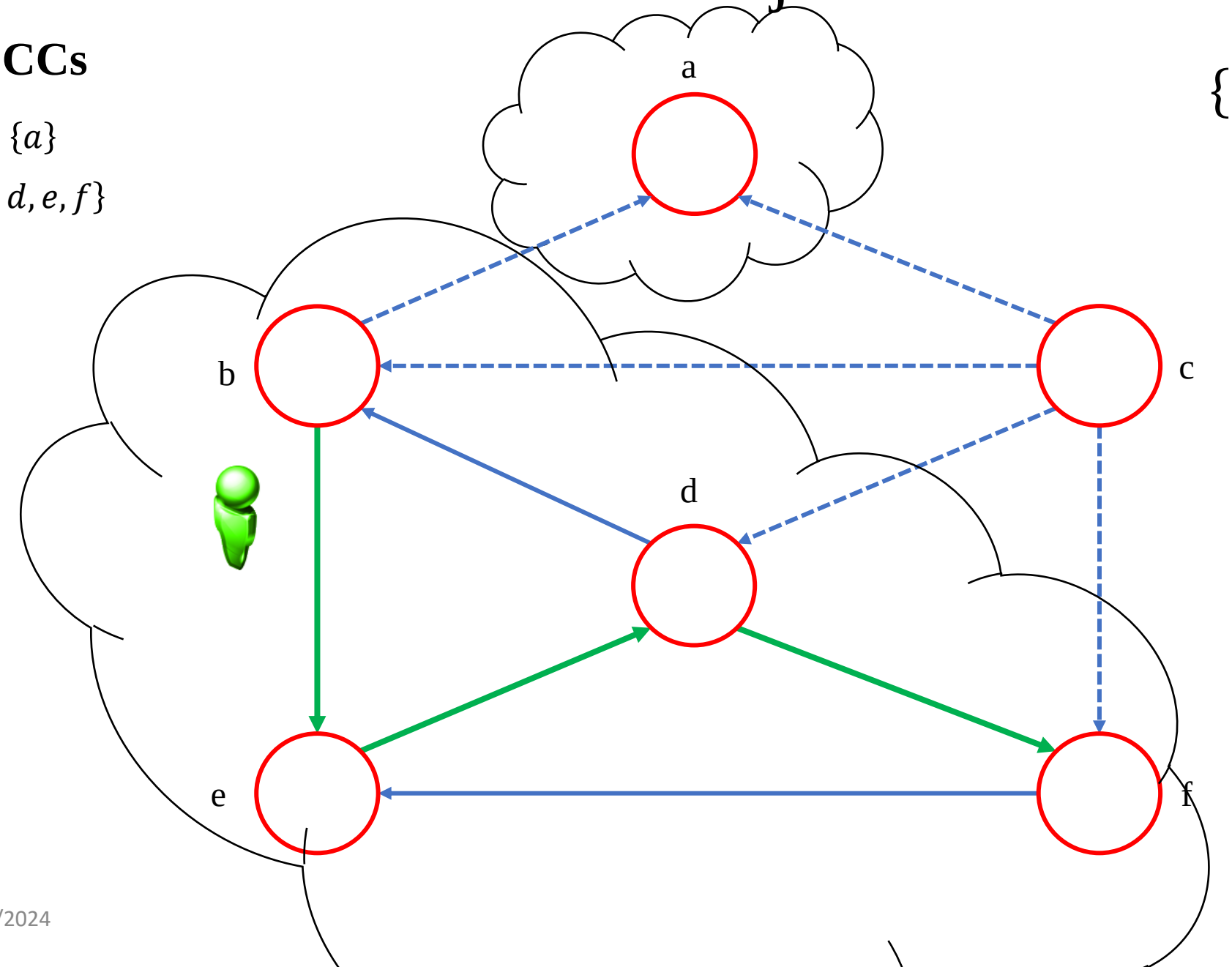
Kosaraju

SCCs

$\{a\}$

$\{b, d, e, f\}$

$\{a, b, d, e, f, c\}$



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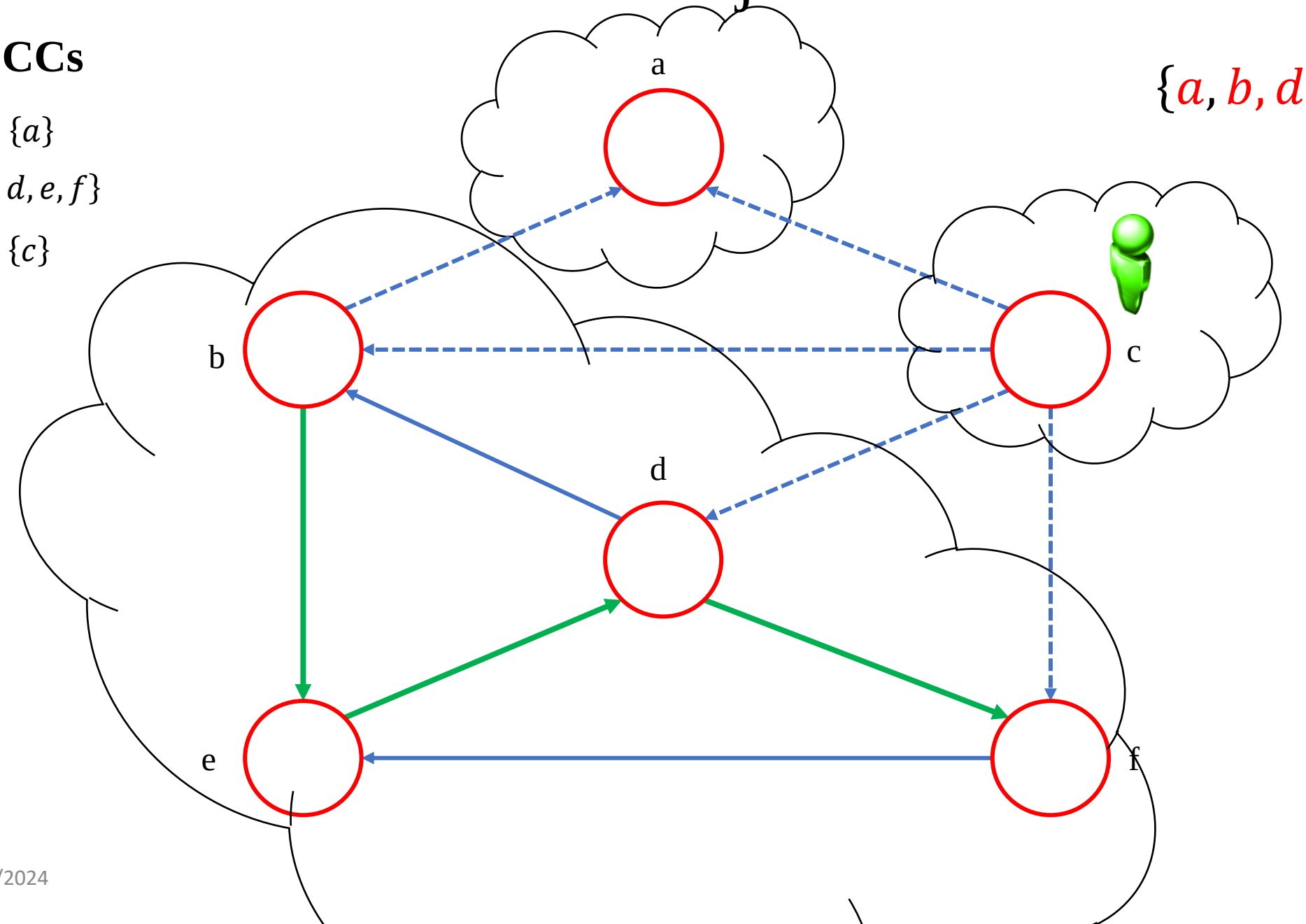
SCCs

$\{a\}$

$\{b, d, e, f\}$

$\{c\}$

$\{a, b, d, e, f, c\}$



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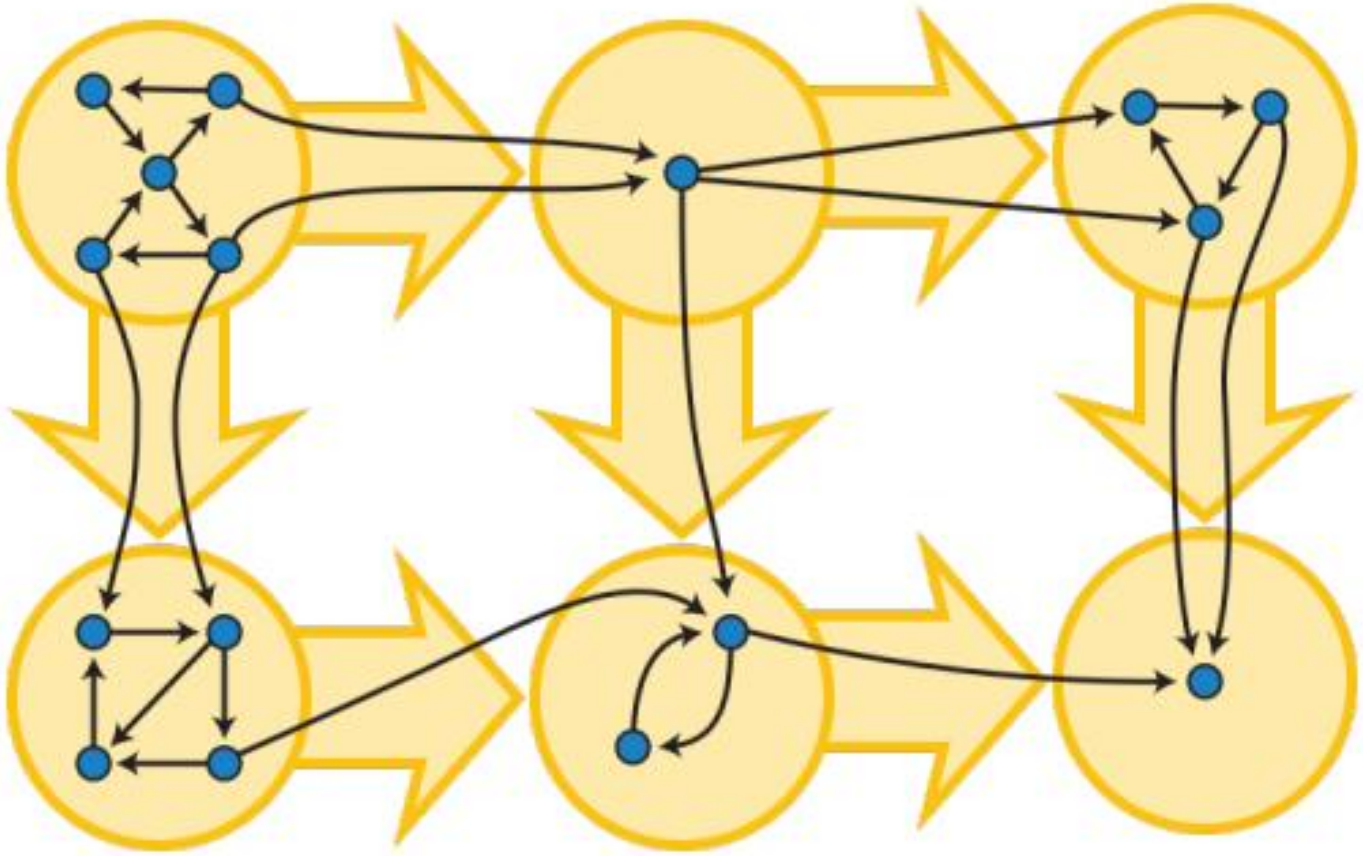
```
vector<vector<int>> adj, adj_rev;
vector<bool> used;
vector<int> order, component;

void dfs1(int v) {
    used[v] = true;

    for (auto u : adj[v])
        if (!used[u])
            dfs1(u);

    order.pb(v);
}

void dfs2(int v) {
    used[v] = true;
    component.pb(v);

    for (auto u : adj_rev[v])
        if (!used[u])
            dfs2(u);
}
```

```
int main() {
    int n,m; cin>>n>>m;
    adj.resize(n); adj_rev.resize(n);
    for(int i = 0; i < m; i++) {
        int a, b; cin>>a>>b;
        a--; b--;
        adj[a].pb(b);
        adj_rev[b].pb(a);
    }
    used.assign(n, false);
    for (int i = 0; i < n; i++){
        if (!used[i])
            dfs1(i);
    }
    used.assign(n, false);
    reverse(order.begin(), order.end());
    for (auto v : order){
        if (!used[v]) {
            dfs2 (v);
            for(int i = 0; i < sz(component); i++) cout<<component[i]+1<<" ";
            cout<<endl;
            component.clear();
        }
    }
}
```

Problema Ejemplo

- https://codeforces.com/problemset/problem/427/C

Condensación



2-SAT

- SAT (Boolean satisfabiality problem) es el problema de asignar valores booleanos a variables con el fin de satisfacer cierta fórmula. En general dicha fórmula se da en forma normal conjuntiva.
- 2-SAT es una restricción de SAT en donde cada clausula tiene exactamente dos literales.

$$(a \vee \neg b) \wedge (\neg a \vee b) \wedge (\neg a \vee \neg b) \wedge (a \vee \neg c)$$

2-SAT

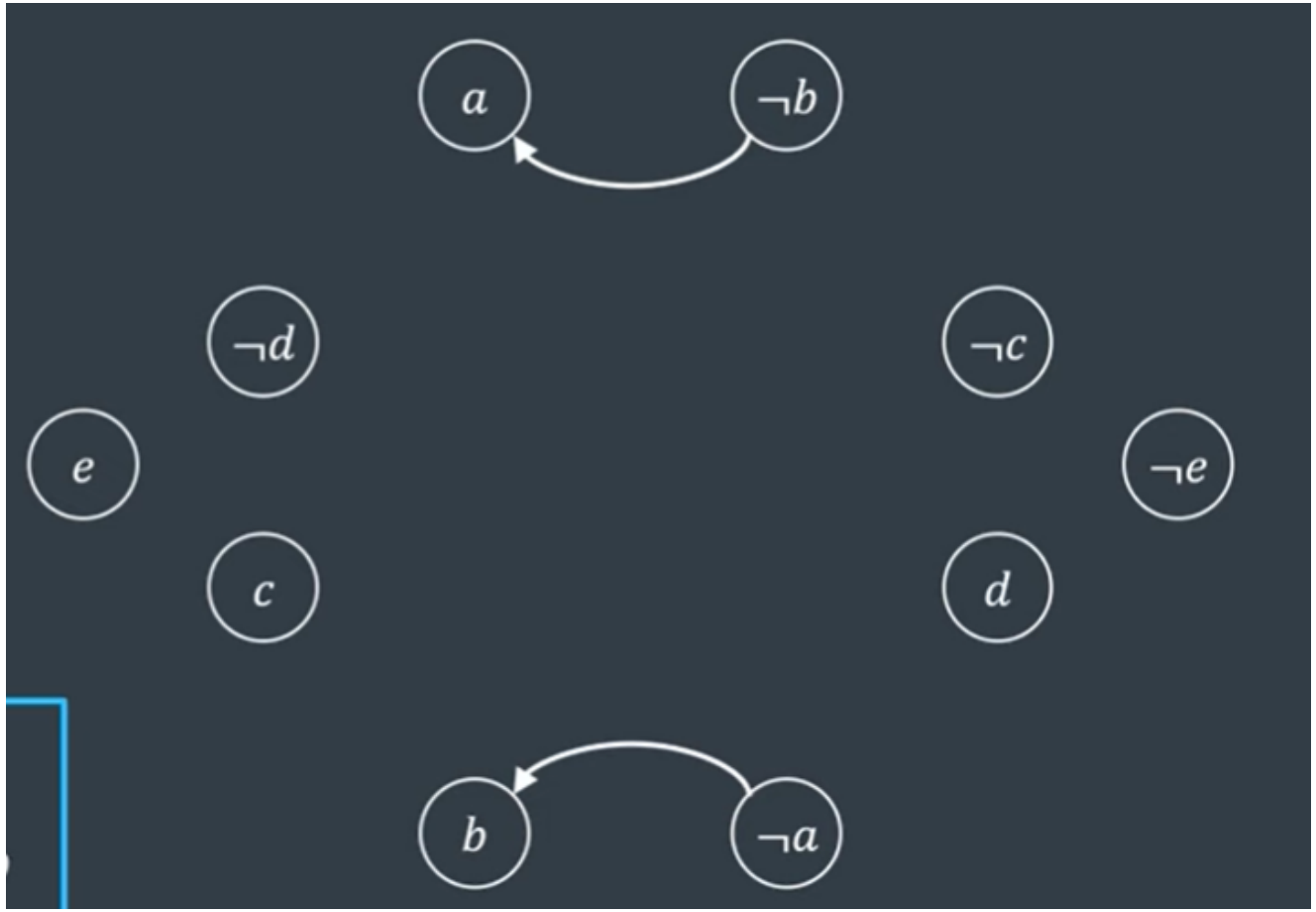
- Notemos que:

$$a \vee b \iff (\neg a \rightarrow b) \wedge (\neg b \rightarrow a)$$

2-SAT

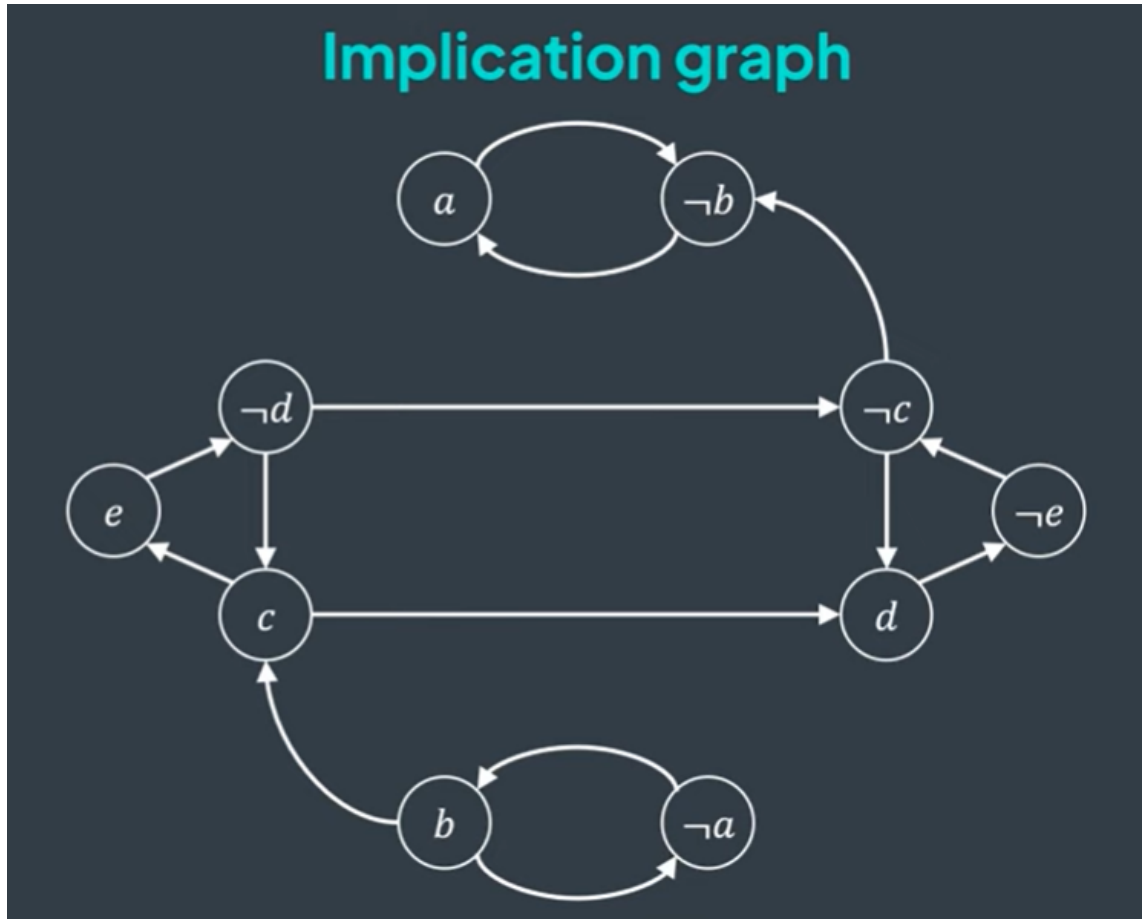
- $a \vee b$
- $\neg b \vee c$
- $\neg a \vee \neg b$
- $c \vee d$
- $\neg c \vee e$
- $\neg d \vee \neg e$
- $\neg c \vee d$

$$a \vee b \Leftrightarrow \neg a \longrightarrow b$$
$$\Leftrightarrow \neg b \longrightarrow a$$

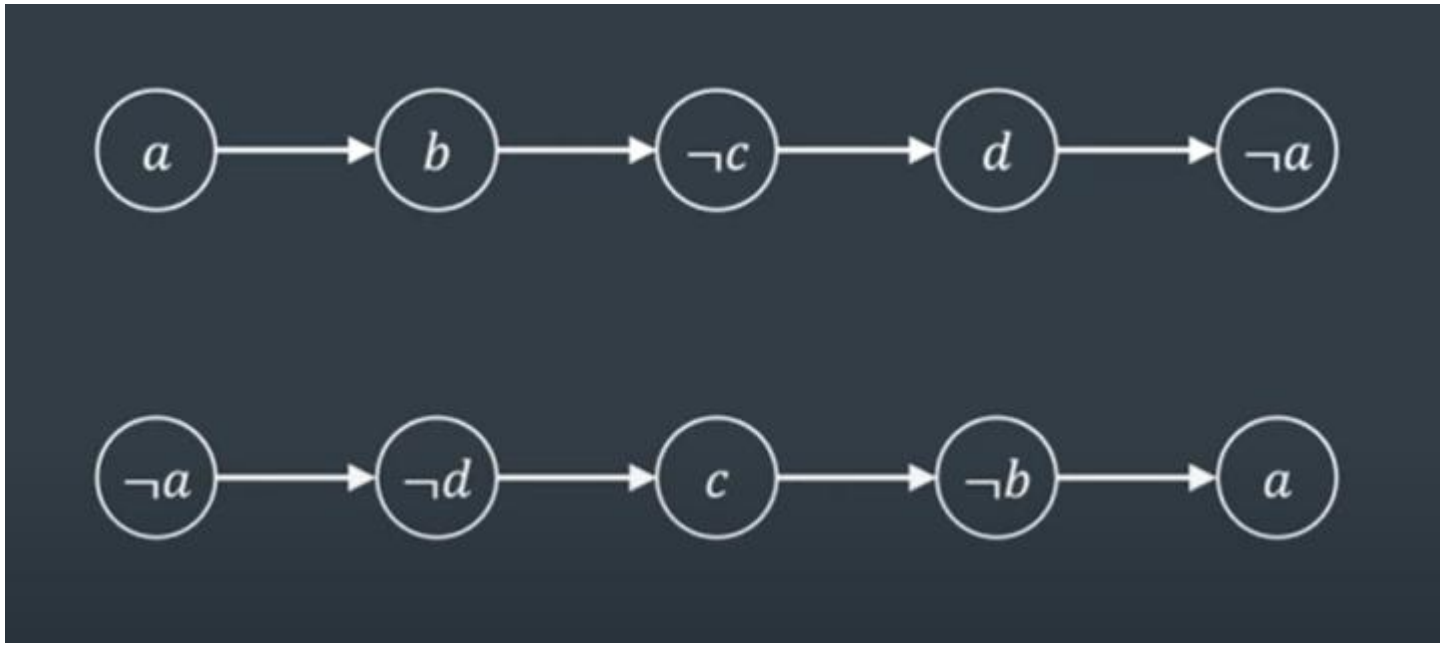


2-SAT

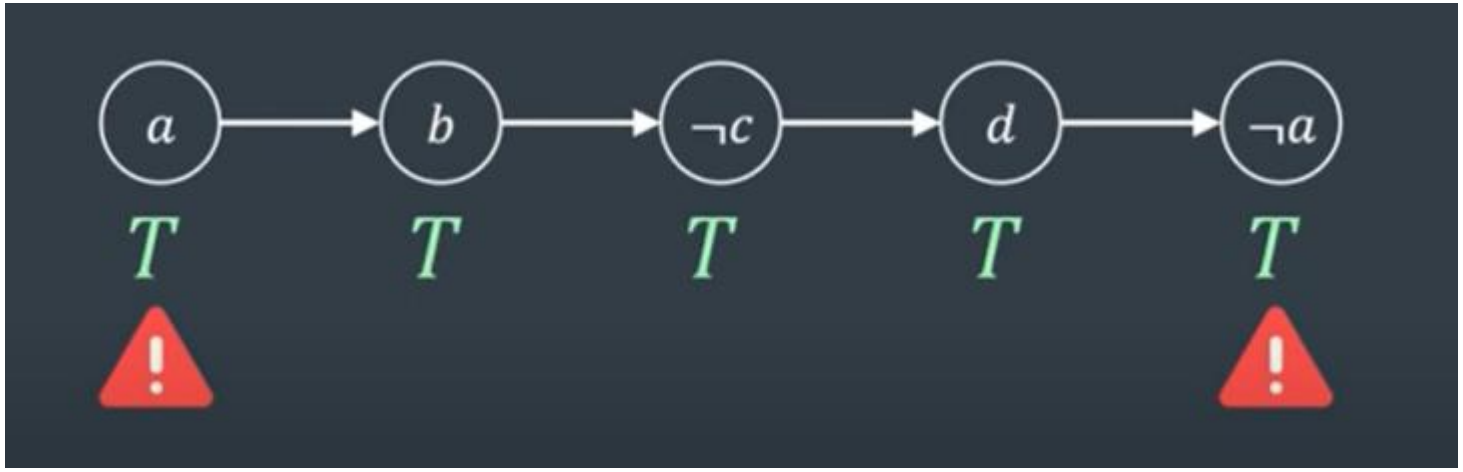
- $a \vee b$
- $\neg b \vee c$
- $\neg a \vee \neg b$
- $c \vee d$
- $\neg c \vee e$
- $\neg d \vee \neg e$
- $\neg c \vee d$

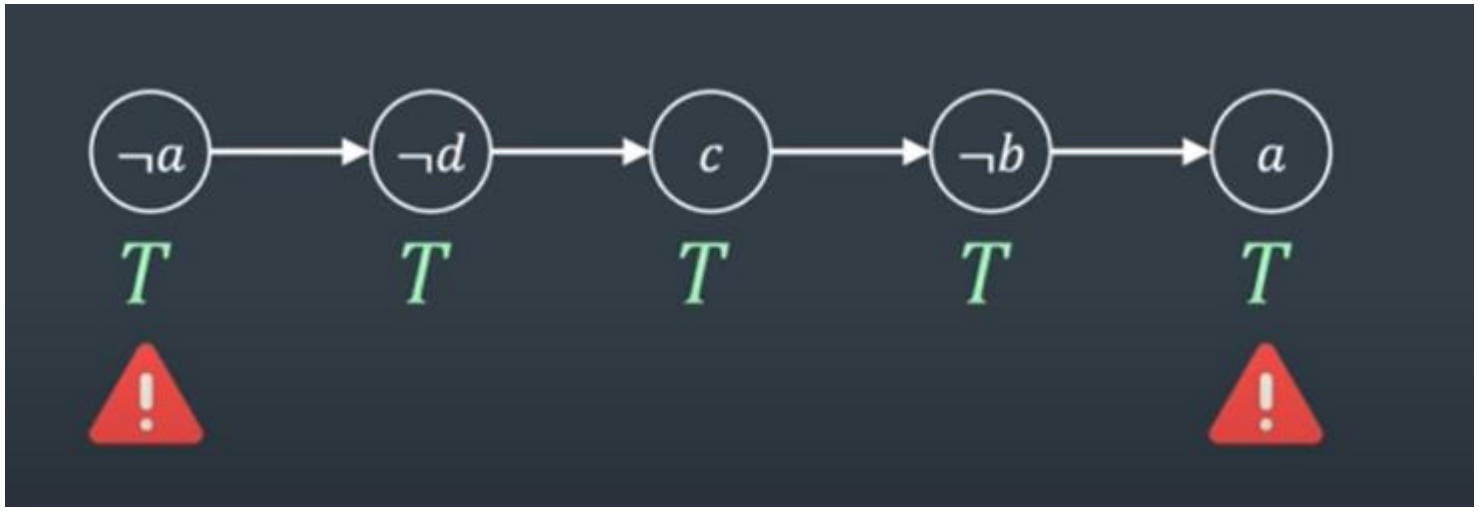


2-SAT



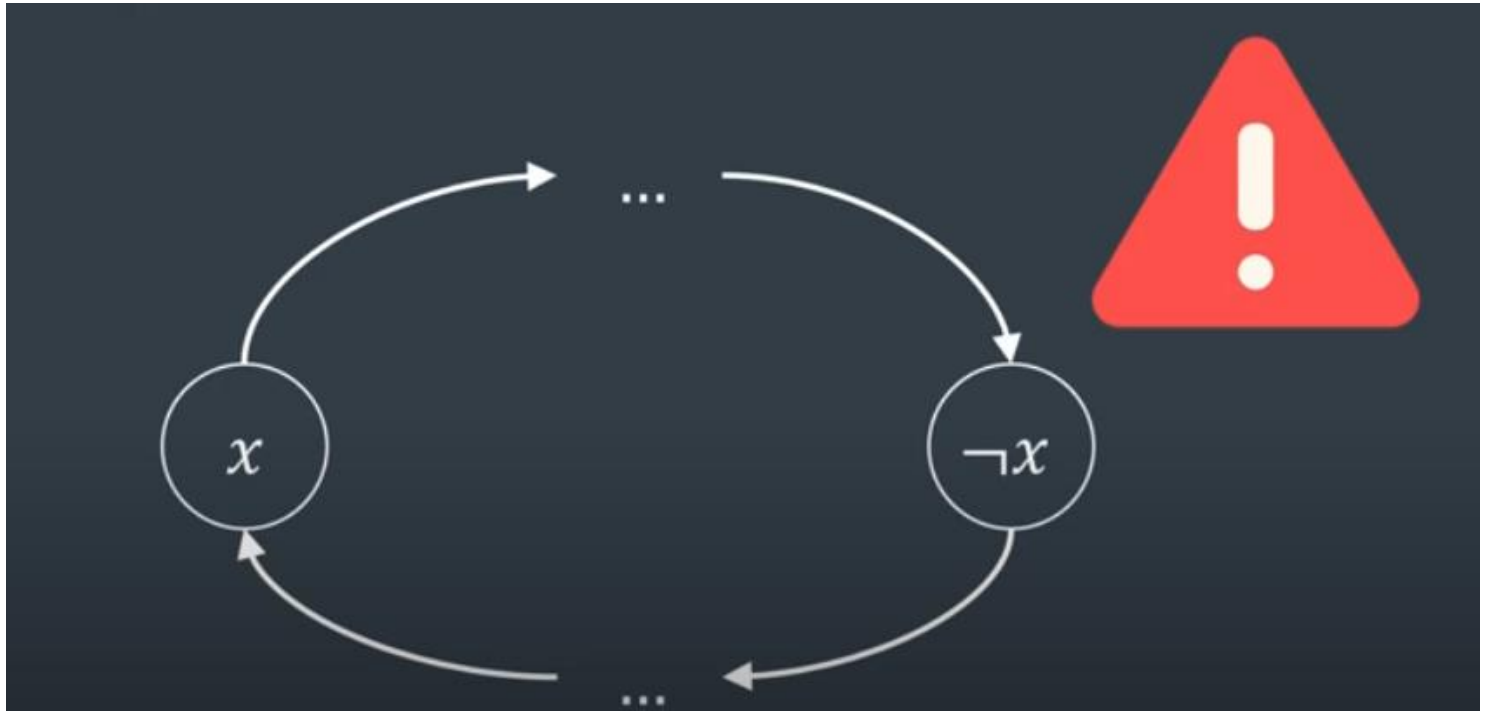
2-SAT





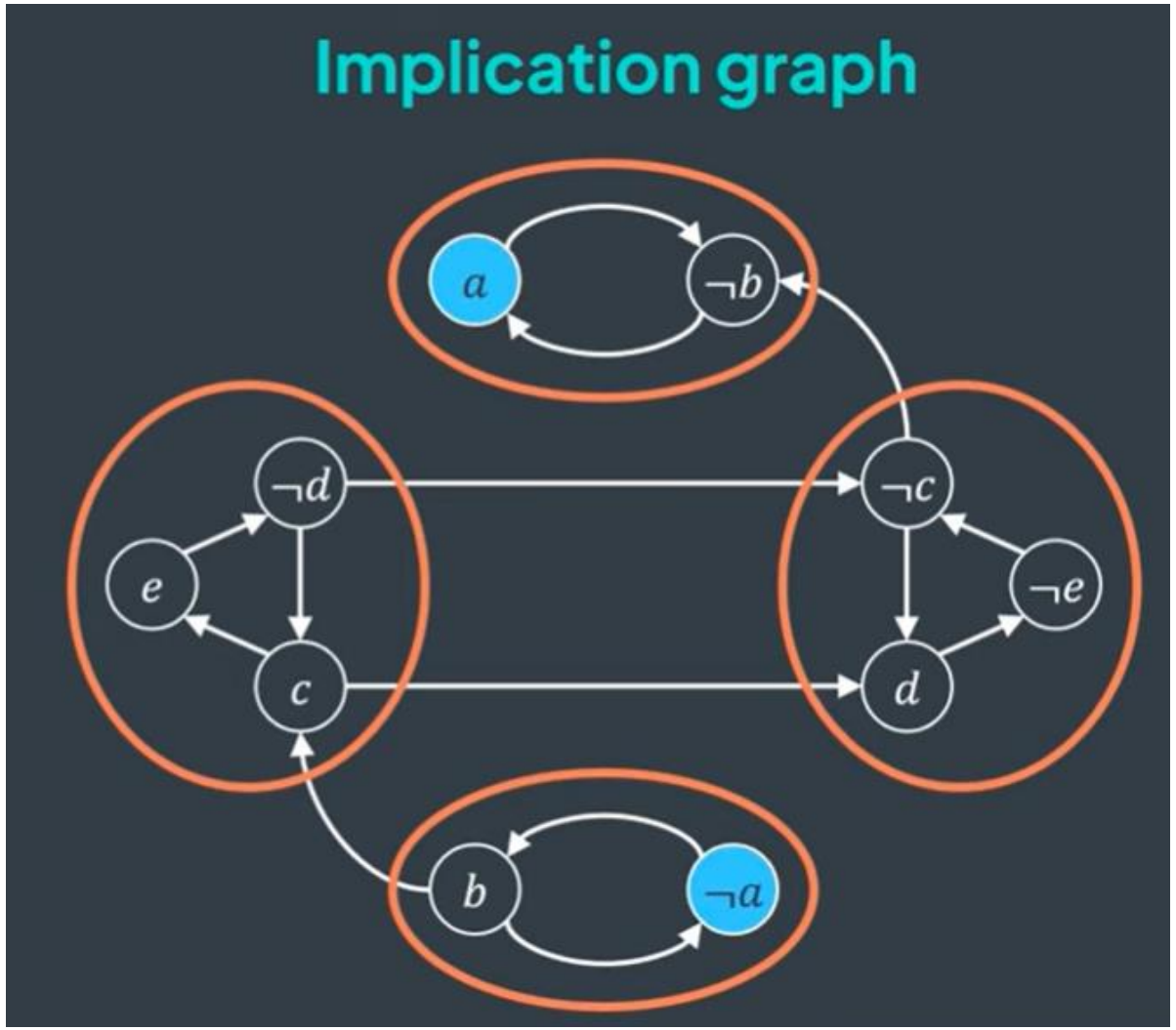
2-SAT

- Si para una variable x, tenemos en el grafo de implicación un camino de x a ¬x y un camino de ¬x a x, la fórmula no es satisfacible.



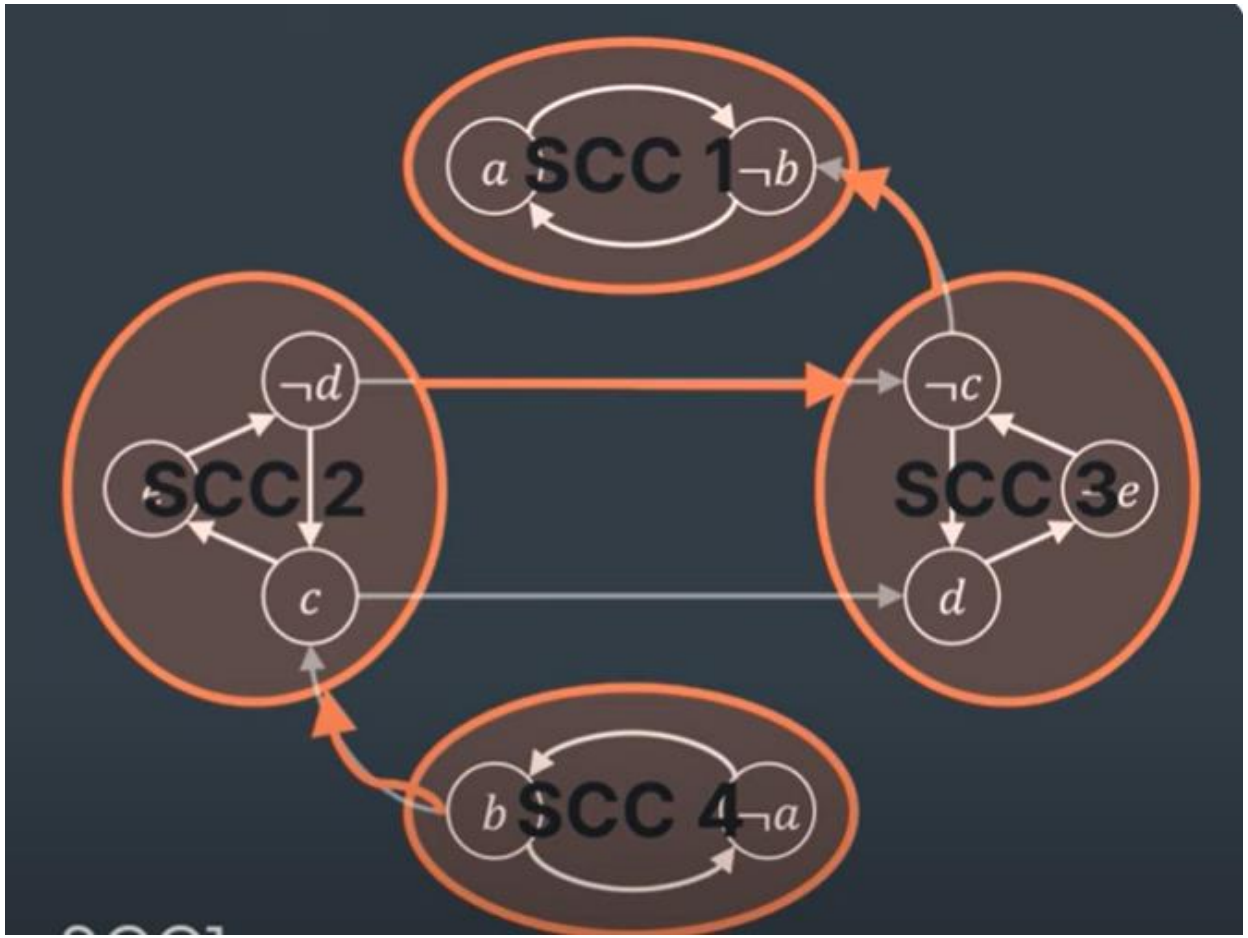
2-SAT

- $a \vee b$
- $\neg b \vee c$
- $\neg a \vee \neg b$
- $c \vee d$
- $\neg c \vee e$
- $\neg d \vee \neg e$
- $\neg c \vee d$



2-SAT

$a = T$
$b = F$
$c = F$
$d = T$
$e = F$



Topological order:
SCC4 – SCC2 – SCC3 – SCC1

2-SAT -> https://cses.fi/problemset/task/1684

```cpp
const ll N = 2e5+10;
char ans[N];
ll n,m,numcomp = 0;

vector<vector<int>> adj(N), adj_rev(N), adj_comp(N), SCC(N);
vector<bool> used(N);
vector<int> order, component, root(N);
ll in[N];

int flip(int x){
    return (x&1)?x+1:x-1;
}

void add_edge(char c1, int a, char c2, int b){
    a = 2*a - (c1 == '-');
    b = 2*b - (c2 == '-');
    adj[flip(a)].pb(b);
    adj[flip(b)].pb(a);
    adj_rev[b].pb(flip(a));
    adj_rev[a].pb(flip(b));
}

void dfs1(int v) {
    used[v] = true;

    for (auto u : adj[v])
        if (!used[u])
            dfs1(u);

    order.pb(v);
}

void dfs2(int v) {
    used[v] = true;
    component.pb(v);

    for (auto u : adj_rev[v])
        if (!used[u])
            dfs2(u);
}
```

```cpp
int main(){
    fast_cin();
    cin>>m>>n;
    for(int i = 0; i < m; i++){
        char c1,c2; ll a,b;
        cin>>c1>>a>>c2>>b;
        add_edge(c1,a,c2,b);
    }
    used.assign(2*n+5, false);
    for (int i = 1; i <= 2*n; i++){
        if (!used[i]){
            dfs1(i);
        }
    }
    used.assign(2*n+5, false);
    reverse(order.begin(), order.end());
    for (auto v : order){
        if (!used[v]){
            dfs2(v);
            numcomp++;
            for(auto x: component){
                root[x] = numcomp;
                adj_comp[numcomp].pb(x);
            }
            component.clear();
        }
    }
    for(int i = 1; i <= n; i++){
        if(root[2*i] == root[2*i-1]){
            cout<<"IMPOSSIBLE"<<endl;
            return 0;
        }
    }

    for(int i = 1; i <= 2*n; i++){
        for(auto x: adj[i]){
            if(root[i] != root[x]){
                SCC[root[i]].pb(root[x]);
                in[root[x]]++;
            }
        }
    }
```

```cpp
queue<int> cola;
order.clear();
for(int i = 1; i <= numcomp; i++){
    if(in[i] == 0){
        order.pb(i);
        cola.push(i);
    }
}

while(!cola.empty()){
    int curr = cola.front();
    cola.pop();
    for(auto x: SCC[curr]){
        in[x]--;
        if(in[x] == 0){
            cola.push(x);
            order.pb(x);
        }
    }
}

used.assign(2*n+5, false);
reverse(order.begin(), order.end());
for (auto v : order){
    for(auto x: adj_comp[v]){
        int i = (x+1)/2;
        if(!used[i]){
            ans[i] = (x&1 ?'-':'+');
            used[i] = true;
        }
    }
}
for(int i = 1; i <= n; i++){
    cout<<ans[i]<<" ";
}
return 0;
```